

Week 6

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

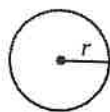
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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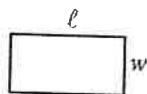
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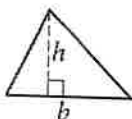


$$A = \pi r^2$$

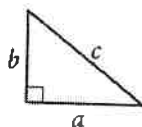
$$C = 2\pi r$$



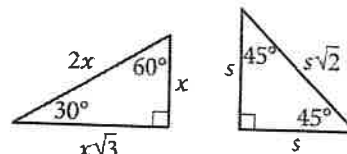
$$A = \ell w$$



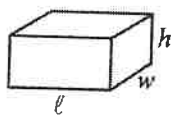
$$A = \frac{1}{2}bh$$



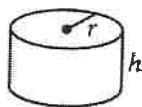
$$c^2 = a^2 + b^2$$



Special Right Triangles



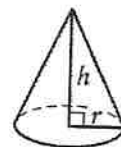
$$V = \ell wh$$



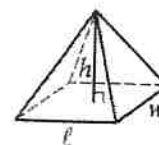
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

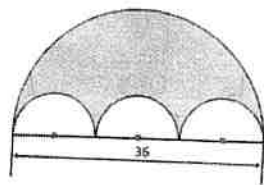
The sum of the measures in degrees of the angles of a triangle is 180.

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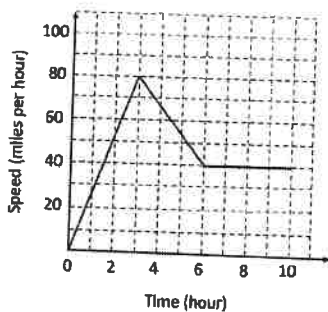
1



In the figure above, three small congruent semi-circles are inscribed to the bigger semi-circle as shown. If the diameter of the bigger semi-circle is 36, what is the value of the area of the shaded region?

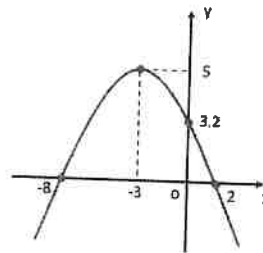
- A) 270π
- B) 216π
- C) 108π
- D) 54π

2



The graph above shows the speed profile for James's car while he was traveling LA to San Francisco. What is the total distance, in miles, traveled for 10 hours trip?

3



Which of the following quadratic forms shows the maximum value of the function as a constant in the equation?

- A) $y = -\frac{1}{5}x^2 - \frac{6}{5}x + \frac{16}{5}$
- B) $y = -\frac{1}{5}(x-2)(x+8)$
- C) $y = -\frac{1}{5}(x+3)^2 + 5$
- D) $y = -\frac{1}{5}(x^2 + 6x) + \frac{16}{5}$

4

If a line is perpendicular to $y = -\frac{1}{2}x + 3$ and passes through $(0, -3)$ in the XY-plane, what is the equation of the line?

- A) $y = -2x - 3$
- B) $y = 2x - 3$
- C) $y = \frac{1}{2}x - 3$
- D) $y = 2x + 3$

5

$$x(3x + 1)^2(x - 1)^3 = 0$$

In the equation above, what is the solution set to the equation?

- A) $\{0, 1, -\frac{1}{3}, \frac{1}{3}\}$
- B) $\{0, 1, -1, -\frac{1}{3}\}$
- C) $\{0, 1, -\frac{1}{3}\}$
- D) $\{0, 1, \frac{1}{3}\}$

7

$$10^x \cdot 1000^{2x} = 100^{4x} \cdot 10$$

In the equation above, what is the value of x ?

- E) -1
- F) 0
- G) 1
- H) 2

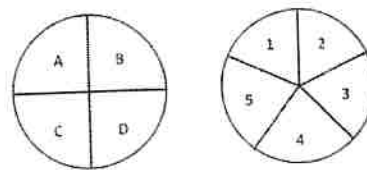
6

$$f(x) = 120,000 \cdot (0.95)^x$$

The function above represent to estimate the value of a certain luxury vehicle, in dollars, where x is the number of years after 2000. Which of the following best interpret the number 0.95 in this context?

- E) The estimated value of the vehicle, in dollars, in 2000.
- F) The estimated percent increase in value each year from 2000.
- G) The estimated percent, in decimal, of the value of the vehicle each year after 2000.
- H) The estimated percent of decrease in total value after 2000.

8



In two dart boards above, assuming that letters and numbers are equally spaced out on both circle boards, what is the probability that a dart will land on C and the other dart will land on even number if two darts are thrown?

- A) $\frac{1}{5}$
- B) $\frac{1}{10}$
- C) $\frac{2}{9}$
- D) $\frac{1}{9}$

9

In the XY -plane, the equations $y = 2x - 5$ and $-4x + ky = -10$ represent the same line. What is the value k ?

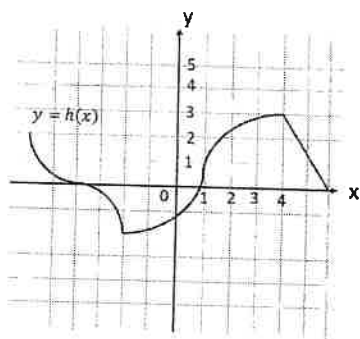
11

$$\frac{2x}{3-x} = \frac{x}{4}$$

In the equation above, which of the following is the solution set to the equation?

- A) $\{0, -5\}$
- B) $\{-5\}$
- C) $\{0, 3\}$
- D) $\{3, -5\}$

10

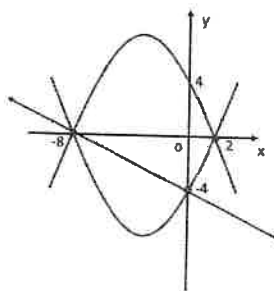


x	$F(x)$
-2	-2
-1	0
0	5
1	8
2	12
3	$1/2$

The complete graph of the function $y = h(x)$ and some values for $F(x)$ are shown in the table above. If the function $h(x)$ has a maximum at $x = k$, what is the value of $F(k - 2)$?

- A) 2
- B) -2
- C) $1/2$
- D) 12

12



A system of 3 equations is shown above in the XY -plane. How many solution(s) does the system have?

- A) Zero
- B) One
- C) Two
- D) Three

13

$$\begin{aligned} y &\leq -2x - 1 \\ -y - 1 &\leq 0 \end{aligned}$$

In the system of inequalities above, which of the following ordered pairs (x, y) satisfies the system?

- I) $(0, 0)$
- J) $(0, 1)$
- K) $(0, -1)$
- L) $(1, 1)$

14

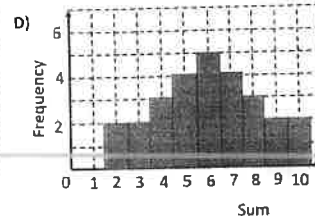
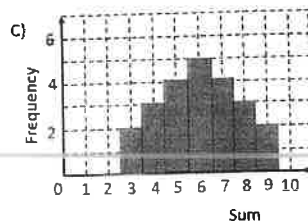
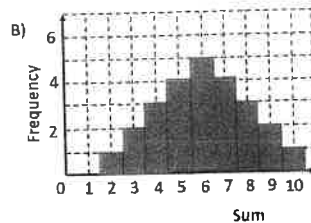
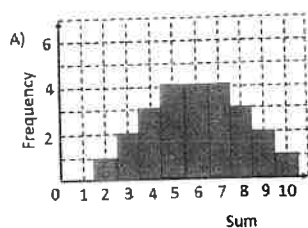
$$\sqrt{2x + 3} + x = 0$$

What is the solution set to the equation above?

- A) $\{3\}$
- B) $\{-1, 3\}$
- C) $\{-1\}$
- D) *No solution*

15

Suppose that you have two spinners and each spinner has numbers 1 through 5, equally spaced out. Which of the following graphs correctly represents for the sum of the numbers in two spinners?



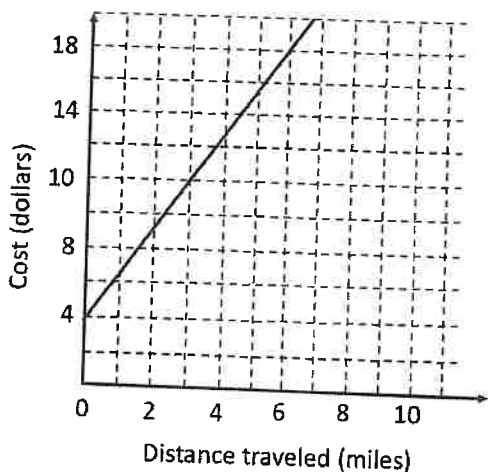
16

Which of the following statements are always true?

- I. $|a - b| = |b - a|$
- II. $-\frac{|a|}{|b|} = \frac{-a}{b}$
- III. $\frac{1}{|a-b|} = \frac{1}{|b-a|}$

- A) I only
- B) I and II only
- C) II only
- D) I and III only

17



The line graphed in the XY -plane above models the total cost, in dollars, for Uber ride, y , in a certain area based on the number of miles in distance traveled, x . Based on the graph, what is the meaning of slope in the line graph?

- A) Total cost, in dollars, for x miles traveled.
- B) Rate of cost, dollars per each additional mile traveled.
- C) Rate of speed which the driver drive.
- D) Rate of cost, dollars per every 2 miles traveled.

18

Sam visited a bridge (Danyang-Kunshan Grand Bridge) in China over the summer and he found out that it is officially the longest bridge in the world which stretches out 164.8 kilometers.

Approximately how many miles is the bridge? (1 kilometer ≈ 0.6214 miles)

- A) 98
- B) 100
- C) 102
- D) 104

19

$$(2 - 3ax)(x + 3) + x^2 - 6$$

In the equation above, a is a constant. If the expression is equal to $-kx$, where k is a constant, what is the value of k ?

20

A manufacturing company bought a robot automation system valued at \$500,000. The marketing department expected that the cost would pay off after some time by saving the cost for employee's labor. If the company saves the same amount, \$70,000 every six months, which of the following equations gives the value of time (t), in year, that the company would take to pay off the purchasing cost?

- A) $y = 500,000 + 70,000t$
- B) $y = -70,000t + 500,000$
- C) $y = 500,000 - 140,000t$
- D) $y = 500,000 + 140,000t$

21

Some radio channel host invited listeners to respond a poll on the show's website that asked, "Do you support the proposition discussed during the show? The show host reported that 80% responded "No" and 18% responded "Yes" on the next session of the show. Which of the following best explains why these results are **unlikely** to represent the sentiments of the entire population?

- A) The response must be 50% "Yes" and 50% "No" to be fair discussion in the show.
- B) The percentage don't add up to 100%, so no conclusion can be made.
- C) Those who responded to the poll were volunteered, not a random sample of the population.
- D) The show host could be against the proposition such that 80% of responders were against it.

22

$$\frac{\frac{9}{x+1} + \frac{9}{x^2+x}}{\frac{x^2}{x+1} - \frac{3x+4}{x+1}}$$

For $x \neq 1, 3$, which of the following expressions is equivalent to the above?

- A) $\frac{9(x+1)}{x(x+4)(x-1)}$
- B) $\frac{9}{x(x-4)}$
- C) $\frac{-9}{x(x-4)}$
- D) $\frac{x+1}{x(x-4)}$

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

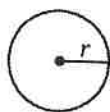
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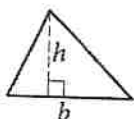


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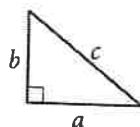
$$C = 2\pi r$$



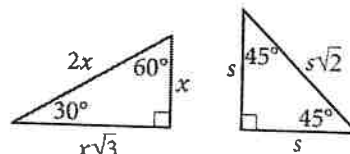
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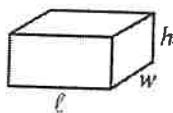
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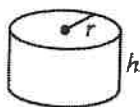
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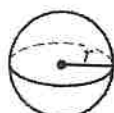
Special Right Triangles



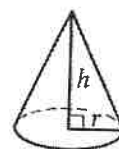
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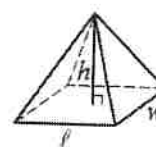
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

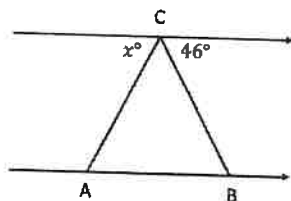
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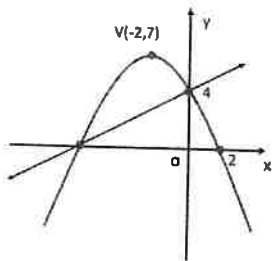
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1



In the figure above, Isosceles $\triangle ABC$ is located between two parallel lines. If $\overline{AC} \cong \overline{BC}$, what is the measure of x° ?

2



The graphs of a linear function and a quadratic function intersect at two points as shown in the XY -plane. The vertex of the parabola is $V(-2, 7)$ and the parabola passes through x axis at $(2, 0)$. What is the equation of the line?

- A) $y = \frac{2}{3}x + 4$
- B) $y = x + 4$
- C) $y = 2x + 4$
- D) $y = \frac{1}{3}x + 4$

3

$$F = G \frac{Mm}{r^2}$$

The equation above shows the formula for the gravitational force between two objects, where G is a gravitational force constant. If r is halved and m is doubled, how does the change affect to the gravitational force, F ?

- A) No effect on F .
- B) F will be doubled.
- C) F will be $\frac{1}{8}$ times.
- D) F will be 8 times.

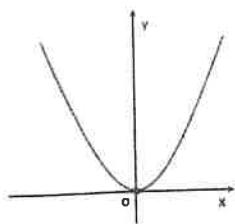
4

$$2x^2 + k + 3kx - 3 = (2x + m)(x + 1)$$

In the equation above, m and k are constants. If the equation is true for all values of x , what is the value of m ?

- A) -0.5
- B) -1
- C) -2.5
- D) -3.5

5



In the quadratic function in the XY -plane above, if the equation of the function is $y = ax^2 + bx + c$, which of the following must be true?

- IV. $a > 0$
- V. $b = 0$
- VI. $c > 0$

- E) I only
- F) I and II only
- G) I and III only
- H) I, II, and III

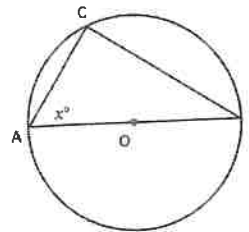
6

$$k^{x^2-4x-1} = \left(\frac{1}{k}\right)^{4x}$$

In the equation above, what is the value of x , where k is a positive constant?

- E) -1
- F) 2
- G) 4
- H) -4

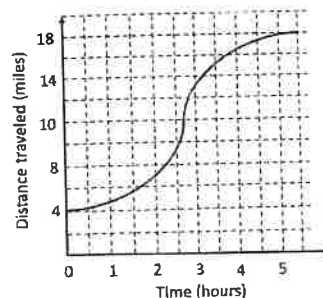
7



In the figure above, triangle ABC is inscribed into a circle O . If the radius of the circle O is 2, which of the following equations correctly represents to find the length of $\overline{AC}$?

- A) $\overline{AC} = 4\sin x^\circ$
- B) $\overline{AC} = 2\cos x^\circ$
- C) $\overline{AC} = 4\cos x^\circ$
- D) $\overline{AC} = 4\tan x^\circ$

8

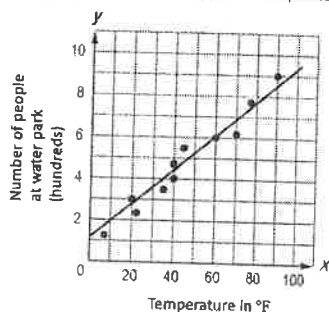


Philip drove from home to Las Vegas. There were some traffic delays because it was a national holiday. In which time interval does the graph show the greatest average rate of change, miles per hour?

- A) Between 1 and 2
- B) Between 2 and 3
- C) Between 3 and 4
- D) Between 4 and 5

9

Number of people at water park vs temperature



The scatter plot above shows the number of people, in hundreds, at an indoor water park and the temperature (°F) in a certain city. A line of best fit is graphed. Which of the following best represents the equation of the line of best fit?

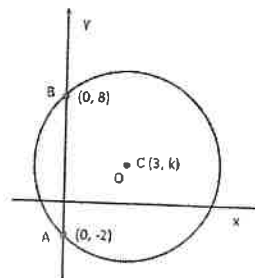
- A) $y = 1.2x + 1.2$
- B) $y = 0.08x + 1.2$
- C) $y = 0.08x + 2$
- D) $y = 1.2x + 2$

10

A wooden crate contains m boxes of paper towels and each box contains n paper towels. If 10 crates cost k dollars, how much is cost per paper towel in cents?

- A) $\frac{100k}{mn}$
- B) $\frac{10mn}{k}$
- C) $\frac{10k}{mn}$
- D) $\frac{10km}{n}$

11



A circle O is graphed in the XY -plane. If the center of the circle is $C(3, k)$ as shown above, where k is a constant, what is the equation of the circle?

- A) $(x - 3)^2 + (y - 3)^2 = 34$
- B) $(x + 3)^2 + (y + 3)^2 = 34$
- C) $(x - 3)^2 + (y - 3)^2 = \sqrt{34}$
- D) $(x + 3)^2 + (y + 3)^2 = \sqrt{34}$

12

A researcher found out that a new system of manufacturing facilities will lose approximately 4% of its value every year. If a new system cost \$55,000 when it was purchased, the value of the system of manufacturing facilities can be modeled by $f(t) = 55,000 \cdot (k)^t$, where k is a constant and t is the number of years after it was purchased. What is the value of k ?

- A) 1.04
- B) 0.96
- C) 1.40
- D) 0.04

13

Kim drove from home to a library at the average speed of 30 mph, and returned home along the same route at the average speed of 50 mph. What is the average speed the entire round-trip, in mph?

14

A quadratic function can be modeled for the height of a ball after thrown from the ground in terms of time, in seconds, after throwing in the air. According to the model, a ball was thrown from the ground and reached the maximum height of 100ft in 5 seconds after it was thrown. How long will it take when the ball hit the ground after it was thrown?

- E) 8 sec
- F) 10 sec
- G) 15 sec
- H) 17 sec

15

Margaux planned for a fundraiser by selling lemonade drinks near her house. She spent \$150 for one-time setup cost and each cup of lemonade costs her \$0.50 and sells each cup for \$2.75. Which of the following equations could represent to find how many cups she would need to sell in order to make a profit of \$100?

- A) $100 = 2.75x - 150$
- B) $100 = 2.50x + 150$
- C) $150 - 2.25x + 100 = 0$
- D) $150 - 100 = 2.25x$

16

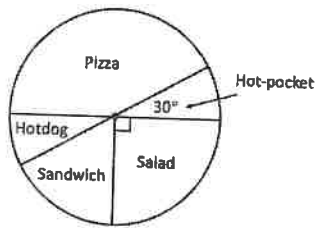
$$\begin{aligned}x^2 + y^2 &\leq 36 \\ x - 3 &\geq 0\end{aligned}$$

In the inequalities above, what is the greatest y value to satisfy the given constraints?

- A) $3\sqrt{3}$
- B) 3
- C) 4
- D) 6

17

Favorite Lunch menu distribution



Mr. Peter, the teacher in a class, surveyed his 90 students for their favorite lunch menu and the results are shown in the circle graph above. How many students picked "Sandwich" for their favorite lunch menu based on the graph?

- A) 30
- B) 25
- C) 20
- D) 15

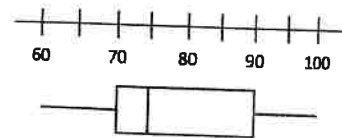
18

When a number, N , is divided by 5, the quotient is q and the remainder is r . Which of the following equations correctly represents N ?

- A) $N = 5r + q$
- B) $N = 5q + r$
- C) $N = qr + 5$
- D) $N = 5r + q$

19

EXAM SCORE DISTRIBUTION



The boxplot represents exam scores for a certain class. Approximately what percent of test scores were between 70 and 75 from the boxplot above?

- A) 5%
- B) 10%
- C) 25%
- D) 50%

20

A number is selected at random from 1 to 40 in the shuffle box. What is the probability that the number is a prime that it is less than 20?

- A) $\frac{1}{5}$
- B) $\frac{9}{40}$
- C) $\frac{8}{20}$
- D) $\frac{7}{40}$

21

Which of the following table values could represent for an exponential decay function?

A)	x	1	2	3	4
	f(x)	27	9	3	1

B)	x	1	2	3	4
	f(x)	1	3	9	27

C)	x	1	3	6	9
	f(x)	10	7	4	1

D)	x	1	3	6	9
	f(x)	1	4	7	10

22

Which of the following expressions are equivalent to $\left(\frac{1}{a} - \frac{1}{b}\right)^2$?

- I. $\left(\frac{1}{a} + \frac{1}{b}\right)^2$
 II. $\frac{1}{a^2} + \frac{1}{b^2} - 2$
 III. $\left(\frac{1}{a} + \frac{1}{b}\right)^2 - \frac{4}{ab}$

- A) I only
 B) I and III only
 C) II only
 D) III only

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

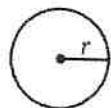
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

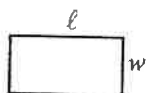
- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

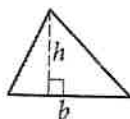


$$A = \pi r^2$$

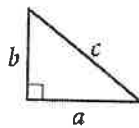
$$C = 2\pi r$$



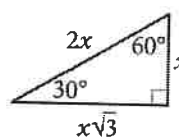
$$A = \ell w$$



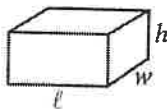
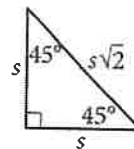
$$A = \frac{1}{2}bh$$



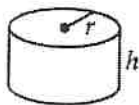
$$c^2 = a^2 + b^2$$



Special Right Triangles



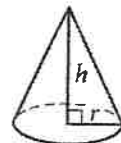
$$V = \ell wh$$



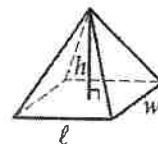
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

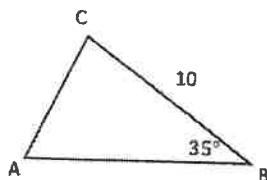
The sum of the measures in degrees of the angles of a triangle is 180.

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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1



Triangle ABC is similar to another triangle FGH (not shown). If $\angle A$ and $\angle C$ correspond to $\angle F$ and $\angle H$, respectively and $AC = \frac{1}{2}FH$, Which of the following must be true?

- I. The perimeter of $\triangle ABC$ is $\frac{1}{2}$ the perimeter of $\triangle FGH$
- II. $m\angle G = 35^\circ$
- III. The length of FG is 20

- A) I only
- B) I and II only
- C) II only
- D) I, II, and III

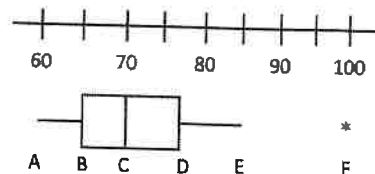
2

A quadratic function is graphed in the XY-plane. Which of the following forms shows where the graph intersects y-axis in the equation?

- A) $y = a(x - h)^2 + k$
- B) $y = a(x - b)(x - c)$
- C) $y - k = a(x^2 + bx)$
- D) $y = ax^2 + bx + c$

3

TEMPERATURE DISTRIBUTION IN A YEAR



In the boxplot above, one of the data (F) is far away from the rest of data, called "outlier". If the outlier is removed from the data, which of the following would change from least to greatest?

- I. Mean
- II. Median
- III. Range

- A) $II < I < III$
- B) $I < II < III$
- C) $III < I < II$
- D) $II < III < I$

4

If a line is parallel to $y = -\frac{1}{2}x - 5$ and has a x-intercept at (4,0) in the XY-plane, what is the equation of the line?

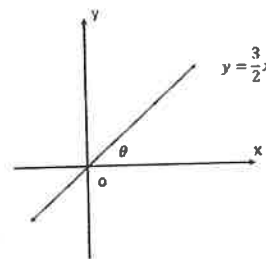
- E) $y = 2x - 8$
- F) $y = 2x - 5$
- G) $y = -\frac{1}{2}x + 2$
- H) $y = -\frac{1}{2}x + 4$

5

$$2x - 5kx = 26$$

In the equation above, k is a constant. If the equation has no solution, what is the value of k ?

7



In the graph above, A line $y = \frac{3}{2}x$ forms an angle θ between the line and positive x -axis in the XY -plane. What is the value of $\sin \theta$?

- A) $\frac{\sqrt{13}}{2}$
- B) $\frac{\sqrt{13}}{3}$
- C) $\frac{3}{2}$
- D) $\frac{3}{\sqrt{13}}$

6

A spinner is divided into 15 equal parts and numbered 1 through 15. What is the probability of spinning a number less than 6 or greater than 10 in a single spin?

- A) $\frac{1}{3}$
- B) $\frac{2}{3}$
- C) $\frac{11}{15}$
- D) $\frac{12}{15}$

8

Two of three sides of a triangle are 10 and 5. Which of the following numbers cannot be a perimeter of the triangle?

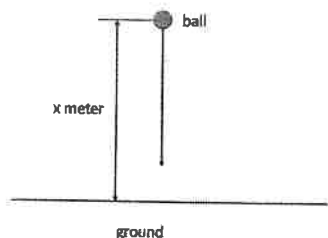
- A) 21
- B) 25
- C) 27
- D) 30

9

Two years ago, Amelie was twice as old as Peter. 6 years ago, Amelie was four times as old as Peter. What is the value of the sum of their current ages?

- A) 20
- B) 21
- C) 22
- D) 24

10



When a ball is released x meter (initial height) above the ground as shown in the figure, the maximum height, in meter, the ball can reach after bouncing once could represent in the following equation.

$$y = \frac{1}{3}x + 2$$

Where x is the initial height, in meters, and x must be greater than 4. If the maximum height, in meters, the ball reaches after bouncing once is half the initial height, what is the initial height the ball should be released?

- A) 2
- B) 6
- C) 12
- D) 24

11

$$y = 2(2a)^{x+1} - a$$

If the exponential function above is graphed in the XY -plane, what is the value of the y -intercept assuming a is a constant?

- A) $-a$
- B) a
- C) $2a$
- D) $3a$

12

An Uber driver charges a basic fare of \$3.25 plus \$0.60 for every $\frac{1}{4}$ of a mile driven. If the total charge for a ride is \$32.05, what is the total distance traveled, in miles?

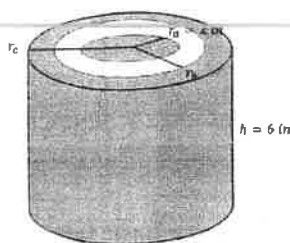
13

$$\begin{aligned} 4 &\leq -2x \\ -y - 1 &\leq 0 \end{aligned}$$

In the system of inequalities above, which of the following ordered pairs (x, y) satisfies the system?

- M) $(0, 0)$
- N) $(-3, -5)$
- O) $(0, 2)$
- P) $(-3, 0)$

14



The cutout of electric wire is shown above. The most inner portion is the place for all electric wires and the white portion of the cylinder is non-conductive material to insulate electricity and the outer most portion is plastic rubber (PVC) to bend freely. If the radius of the inner most portion, r_a , is 2 mm and $r_c = 2r_b = 3r_a$ in the figure, what is the ratio of volumes of white portions to the entire volume?

- A) $\frac{5}{36}$
- B) $\frac{8}{9}$
- C) $\frac{1}{9}$
- D) $\frac{1}{4}$

15

Summer Class Enrollments in ABC Community College

Class Enrollments	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Calculus	23	12	8	10	6	12	2	73
Physics	45	6	7	20	8	11	7	134

The table above shows summer class enrollment distribution during the registration week in ABC Community College. Based on the data, by how much does the median number of enrollments in Calculus class exceed the median number of enrollments in Physics class during the 7 days?

16

In the XY -plane, the graph of $y = x^2 + ax + b$, where a and b are constants, has two x -intercepts at $(2, 0)$ and $(-4, 0)$. What is the sum of $a + b$?

- A) -10
- B) 10
- C) -6
- D) 6

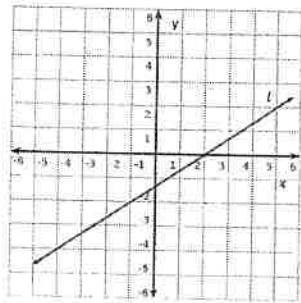
17

$$-3x^2 + 6x - 3y^2 - 3y - 1 = 0$$

In the circle equation above, what is the coordinates of the center of the circle?

- A) $(-1, \frac{1}{2})$
- B) $(1, \frac{1}{2})$
- C) $(1, -\frac{1}{2})$
- D) $(1, -1)$

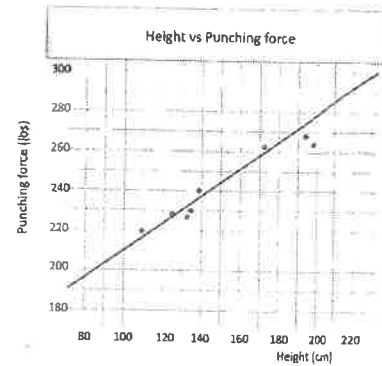
18



In the XY-plane above, line l is shown above. If another line m (not shown) is perpendicular to the line l , which of the following could be the equation of line m ?

- A) $2x + 3y = 2$
- B) $3x + 2y = -1$
- C) $2x - 3y = 9$
- D) $3x - 2y = 3$

19



The scatter plot above shows the punching force, in lbs., and the height of the person, in cm, of 8 people as well as the line of best fit for the data. Which of the following best represents an equation of the line of best fit?

- A) $y = 0.67x + 190$
- B) $y = 0.67x + 143$
- C) $y = 1.5x + 190$
- D) $y = 1.5x + 143$

20

$$(2ax + 3)(x - 1) - x^2 + 3$$

In the equation above, a is a constant. If the expression is equal to mx , where m is a constant, what is the value of m ?

21

After math final exam, a math teacher found out that mean score for class A is at least 20 points lower than class B. If the mean score for class B is y and the mean score for class A is x , which of the following correctly represents this situation?

- A) $y \geq x + 20$
- B) $y \leq x + 20$
- C) $y = x + 20$
- D) $y \leq x - 20$

22

Edison, an electric power company, charges $\$k$ for monthly basic fee plus $\$0.07$ per kilowatt-hour (kWh) of electricity in a certain area. If Philip paid $\$123.14$ for the electric bill for a certain month, which of the following expressions could be used to find how many kilowatt-hours (kWh) of electricity did he use for that month?

- A) $\frac{123.14 - k}{0.07}$
- B) $123.14 - k - 0.07$
- C) $\frac{123.14 - 0.07}{k}$
- D) $\frac{123.14 - 100k}{0.07}$

STOP

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Math

22 QUESTIONS
(TIME: 35 MIN)

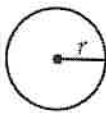
DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

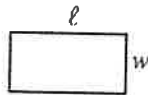
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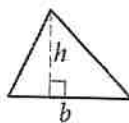
REFERENCE

$$A = \pi r^2$$

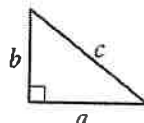
$$C = 2\pi r$$



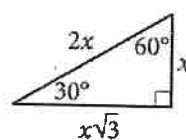
$$A = \ell w$$



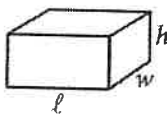
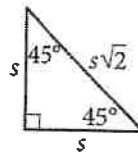
$$A = \frac{1}{2}bh$$



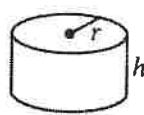
$$c^2 = a^2 + b^2$$



Special Right Triangles



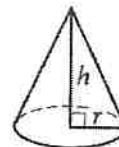
$$V = \ell wh$$



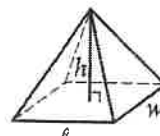
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

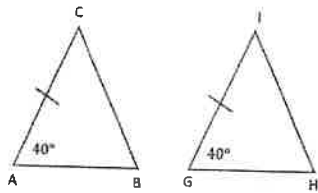
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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1



In triangle ABC and triangle GHI, $m\angle A = m\angle G = 40^\circ$ and $\overline{AC} \cong \overline{GI}$. Which of the following information is sufficient to prove that the triangles are congruent?

- I. $\overline{AC} \cong \overline{HI}$
- II. $m\angle B = m\angle H$
- III. $\angle C \cong \angle I$

- A) I and II
- B) I and III
- C) II and III
- D) I, II, and III

2

$$y = a(b)^x + k$$

In the exponential function above, a , b , and k are constants. What is the y-intercept of the graph of the function in the XY-plane?

- A) $(0, ab + k)$
- B) $(0, k)$
- C) $(0, a + k)$
- D) $(0, b + k)$

3

$$T = 2\pi \sqrt{\frac{l}{g}}$$

The equation above shows the formula for the period of the pendulum (T), where g is a gravitational acceleration. If g is halved in a certain planet and l , the length of the pendulum, is doubled, how does the change affect to the period of the pendulum, T ?

- E) No effect on T .
- F) T will be doubled.
- G) T will be $\frac{1}{2}$ times.
- H) T will be 4 times.

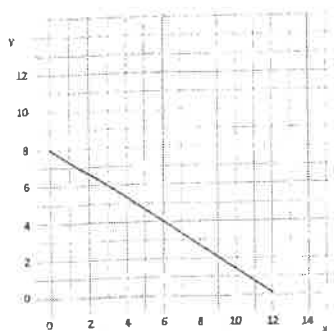
4

Value	Frequency
10	2
12	3
18	1
22	4
25	7

Which of the following statements best explains for the mean and median values from the frequency table above?

- A) The mean is equal to the median.
- B) The mean is greater than the median.
- C) The median is greater than the mean.
- D) Not enough information to calculate.

5



Jonathan sells tickets for music concert and play in theater. The graph above models the possible number of tickets for both music concerts and play. Total amount he collected for tickets is \$240. If he sold a fixed amount of each ticket for music concert and another fixed amount of each ticket for the play, which of the following equations best represent this situation?

- A) $20x + 30y = 240$
- B) $30x + 20y = 240$
- C) $8x + 12y = 240$
- D) $12x + 8y = 240$

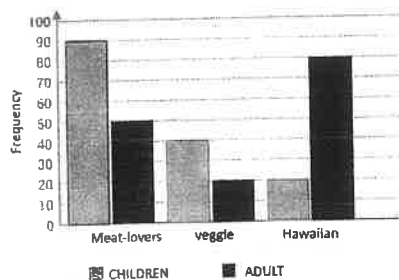
6

$$\left(\frac{1}{a}\right)^{x^2+2x-1} = a^{-2}$$

In the equation above, what is the sum of x values, where a is a positive constant?

- I) -1
- J) 2
- K) 4
- L) -2

7



The bar graph above summarizes the distribution of favorite topping of pizza for both adults and children during one week at a certain Pizza restaurant. What is the ratio of the adults' percent for meat-lover favorite to the children's percent for meat-lover favorite?

- A) 9:5
- B) 5:9
- C) 9:4
- D) 4:9

8

In the XY -plane, the graph of quadratic function $y = ax^2 + bx + c$ has a vertex at $(2, -3)$ and the graph passes through two points on the x -axis. Which of the following can NOT be the value of $a + b + c$?

- A) -3
- B) -2
- C) 0
- D) 2

9

$$\begin{aligned}x + 3y &= -7 \\ -15y &= 5x + 35\end{aligned}$$

In the system of equations above, for a real number k , which of the following points could lie on the graph of the given system of equations in the XY -plane?

- A) $(k, -\frac{1}{3}k - \frac{7}{3})$
- B) $(k, \frac{1}{3}k + \frac{7}{3})$
- C) $(3k - 7, k)$
- D) $(k, -3k - 7)$

10

$$x^2 + 16x - k = 0$$

In the equation above, k is a constant. If the equation has two real solutions, where $m < k$, what is the greatest value of m ?

- A) -63
- B) -64
- C) -65
- D) -66

11

An isosceles right triangle ABC has a perimeter of $16 + 16\sqrt{2}$ cm. What is the length, in inches, of a leg of the triangle?

- A) 8
- B) 16
- C) $8\sqrt{2}$
- D) $16\sqrt{2}$

12

A ball is dropped from a height of 15 meters above the ground and hit the ground several times. Assuming that the ball bounces off $\frac{2}{3}$ its previous height, which of the followings would approximate the maximum height, in meters, of the ball between the second and third time it hit the ground?

- A) 10m
- B) 6.7m
- C) 2.2m
- D) 1.5m

13

To determine if turmeric reduces the risk of type 2 diabetes for men, study was conducted by interviewing random sample of 20,000 men who had no history of diabetes. Participants reported they were turmeric users in their diet. Seven years later, participants were interviewed again and the researcher found out that the proportion of men who diagnosed type 2 diabetes was significantly lowered for men identified as regular turmeric users. Which of the following statements is the most appropriate conclusion of this study?

- A) This study shows that turmeric is a medical cure of type 2 diabetes to anyone.
- B) Turmeric definitely caused the reduction in the risk of type 2 diabetes.
- C) There is an association between turmeric use and the risk of type 2 diabetes for men, but it is not necessarily a cause-and-effect relationship, and the association may not exist for women.
- D) There is an association between turmeric use and the risk of type 2 diabetes for both men and women, but it is not necessarily a cause-and-effect relationship.

14

$$\begin{aligned}x^2 + y^2 &\leq 81 \\ y - 7 &\geq 0\end{aligned}$$

In the inequalities above, what is the least x value to satisfy the given constraints?

- E) 7
- F) -7
- G) $4\sqrt{2}$
- H) $-4\sqrt{2}$

15

A quadratic function can be modeled for Projected Annual Revenue (f), in millions of dollars, based on the unit price in dollars, x , of the products as follows.

$$f(x) = a(x - 1,200)^2 + 20$$

If the quadratic function is open-downward parabola, which of the following best represents the number 1,200 in the context?

- A) The company could approximate a maximum annual revenue 20 million dollars if the price per unit is \$1,200.
- B) The company could approximate a maximum annual revenue 20 million dollars if the company makes 1,200 product units per year.
- C) The company could approximate a maximum revenue 20 million dollars as long as the company sells 1,200 units of their products.
- D) The company could approximate a maximum revenue 1,200 million dollars if the price per unit is \$20.

16

$$D = 2,777 - 505t$$

The equation above shows the estimated distance (D), in miles, from Los Angeles airport (LAX) of the airplane t hours after taking off from New York airport (JFK). Which of the following statement best represents the number, 505, in the context?

- A) Average distance, in miles, the airplane flies every hour.
- B) Average speed, in mph, of the airplane for the entire trip.
- C) The time, in minutes, it takes the airplane from JFK to LAX.
- D) Total distance, in miles, between JFK and LAX.

17

A scientist observes the population of bees in a hive for his project. At the beginning of his observation, there were 12 bees visible in the hive and he found out that the population of bees in the same hive quadruples every week. Which of the following equations best models the number, f , of bees x months after the beginning of his observation?

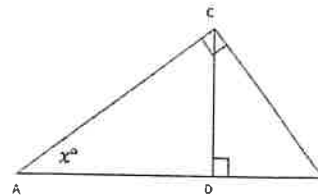
- A) $f = 12(1 + 4x)$
- B) $f = 12(4)^x$
- C) $f = 12(4)^{\frac{x}{4}}$
- D) $f = 12(4)^{4x}$

18

Sam harvested 60,000 pounds of barley last year in his farm. He increased the harvest by 20% this year by applying new method of irrigation system. And he expects another increment to the next year at the same rate as this year by expanding the area of his farm. How many pounds of barley does he expect to harvest in his farm for the next year?

- A) 72,000
- B) 84,000
- C) 86,400
- D) 96,000

19



In the figure above, $\sin x^\circ = 0.75$ and the lengths of AC and BD are 10 inches and 7 inches, respectively. What is the length of BC, in inches?

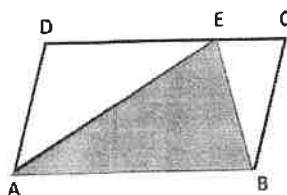
- A) 7.00
- B) 10.26
- C) 9.33
- D) 7.50

20

The area of a forest is decreasing at a rate of 1.2 acre per year in a certain developing country. If 1 square mile is equivalent to 640 acres and 1 kilometer is equivalent to 0.62 miles, which of the following converts the rate correctly which the area of forest is decreasing, in square kilometers per year?

- A) 0.0049
- B) 0.0030
- C) 1997.92
- D) 0.0015

21



In the figure above, point E is located on the side CD of the parallelogram. If $DE:EC = 4:1$ and the area of shaded region is 20, what is the area of triangle BCE?

- A) 3
- B) 4
- C) 5
- D) 6

22

Amount of money spend by each customer

Number of customers	Amount of money spent in dollars
5	Above 300
25	200-300
134	100-200
17	Below 100

The table shows four separated ranges of money, in dollars, spent by each customer in ABC whole sale market during the independence sale. Which of the following can be obtained from the table given?

- I. Average amount of money, in dollars each customer spent.
- II. The median amount of money, in dollars for total 181 customers.
- III. The range of money, in dollars, spent by more than half of all customers during the sale.

- A) I only
- B) I and II only
- C) III only
- D) I, II, and III

STOP

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Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

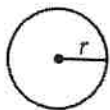
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Use of a calculator is permitted for all questions.

NOTES

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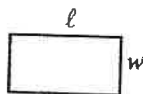
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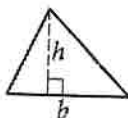


$$A = \pi r^2$$

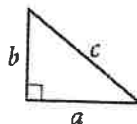
$$C = 2\pi r$$



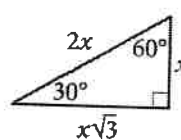
$$A = \ell w$$



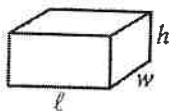
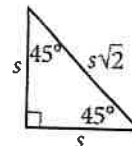
$$A = \frac{1}{2}bh$$



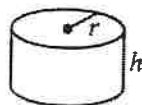
$$c^2 = a^2 + b^2$$



Special Right Triangles



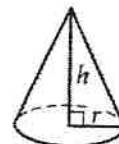
$$V = \ell wh$$



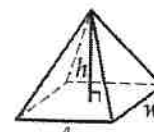
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

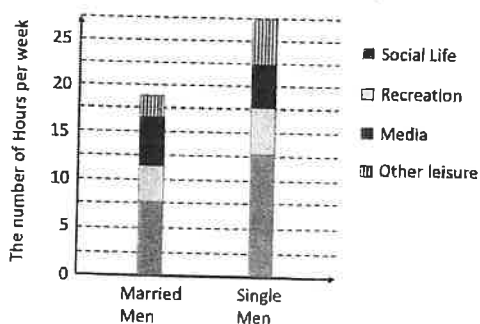
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1

Average number of hours weekly spent in leisure time

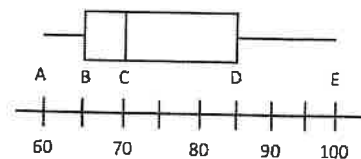


In the bar graph above, which of the following statements interprets correctly?

- I. The percent of weekly time spent on Media for both married men and single men are same.
- II. The percent of weekly time spent on Recreation for Married Men is higher than that of single men.
- III. The number of weekly hours spent on recreation, media, and other leisure for married men is lower than the number of weekly hours spent on social life for single men.

- A) I only
- B) I and II only
- C) II only
- D) II and III

2



Calculus final exam score distribution

In the boxplot above, which of the followings are valid statements?

- I. There must be more data between C and D than between B and C.
- II. There are approximately 50% of data located A and C.
- III. There are approximately 50% of data located between B and D.

- A) I only
- B) I and III only
- C) II and III only
- D) I, II, and III

3

Which of the following expressions is equivalent to

$$(\sqrt{4x} + \sqrt{16y})^{\frac{2}{5}}, \text{ where } x, y > 0?$$

- A) $\sqrt[5]{4x + 16y}$
- B) $(4x + 16y)^5$
- C) $\sqrt[5]{4x + 16y + 16\sqrt{xy}}$
- D) $\sqrt[5]{4x + 16y + 8\sqrt{xy}}$

4

The function f is defined by $f(x) = -2(3)^{\frac{x}{5}} - 6$. What is the coordinate of y -intercept of the graph it is graphed in the XY -plane?

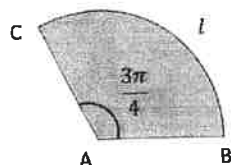
- A) $(-12, 0)$
- B) $(-8, 0)$
- C) $(0, -6)$
- D) $(0, -8)$

6

Susan plans her birthday party and orders for foods. There will be no more than 50 people in the party. The food catering restaurant charges \$25 per adult and \$20 per child and her budget is limited to \$1,200. If the number of adults is a and the number of children is c and no other charges, which of the following systems best represents all constraints mentioned above?

- A) $\begin{cases} a + c \leq 50 \\ 25a + 20c \leq 1,200 \end{cases}$
- B) $\begin{cases} a + c > 50 \\ 25a + 20c \leq 1,200 \end{cases}$
- C) $\begin{cases} a + c \leq 50 \\ 20a + 25c \leq 1,200 \end{cases}$
- D) $\begin{cases} a + c < 50 \\ 25a + 20c > 1,200 \end{cases}$

5



In the sector ABC of a circle above, if the central angle CAB is $\frac{3}{4}\pi$ as shown, what is the fraction of the length of an arc (l) to the entire circumference of the circle A ?

- A) $\frac{13}{32}$
- B) $\frac{3}{8}$
- C) $\frac{1}{3}$
- D) $\frac{7}{16}$

7

If $x + y = 17$ and $xy = 24$, what is the value of $x^2 + y^2$?

- A) 265
- B) 241
- C) 313
- D) 337

8

The average (arithmetic mean) of five numbers is 12. After one of the numbers is removed, the average of the remaining numbers becomes 13.5. What number has been removed?

- A) 6
- B) 8
- C) 10
- D) 12

10

If the ratio of the measures of angles in a triangle is 4:5:6, what is the measure of the largest angle, in degrees, in the triangle?

- A) 48°
- B) 60°
- C) 72°
- D) 120°

9

If sixty eight percent of the people who made it to the final round in a certain competition are female, what is the ratio of the number of males who made it to the final round to the number of females who made it to the final round?

- A) $\frac{8}{25}$
- B) $\frac{17}{25}$
- C) $\frac{8}{17}$
- D) $\frac{25}{8}$

11

$$\begin{aligned}y &< -4x - 2 \\y &< -\frac{1}{4}x - 2\end{aligned}$$

In the system of inequalities above, which of the following ordered pairs (x, y) satisfies the system?

- Q) $(0, 0)$
- R) $(-3, -1)$
- S) $(-1, -5)$
- T) $(3, 0)$

12

A wooden stick of length l inches is cut into two pieces such that the length of one piece is 2 inches more than twice the length of the other piece. Which of the following is the length, in inches, of the longer piece in term of l ?

- A) $\frac{2+l}{3}$
- B) $\frac{2+2l}{3}$
- C) $\frac{1+l}{3}$
- D) $\frac{2l+1}{3}$

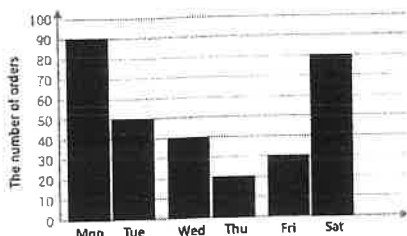
14

$$(2 - x)^2 - (2 - x) + 11 = 0$$

In the equation above, how many distinct real solutions does the equation have?

- A) Zero
- B) One
- C) Two
- D) Infinitely many

13



The bar graph shows the number of orders per day on a local pick-up restaurant last week. For these six days, how much greater is the mean number of orders per day than the median number of orders per day?

- A) 31.67
- B) 21.67
- C) 11.67
- D) 6.67

15

$$-x^2 + 6x - y^2 - 4y - k = 0$$

In the circle equation above, what is the value of k in the equation to have 7 as a radius of the circle?

- A) -36
- B) 36
- C) 6
- D) -6

16

$$Q = mc\Delta T$$

The equation above shows how much energy (Q), in joules, required to raise the temperature of water from T_1 to T_2 (ΔT), in degree Celsius, of m grams of water, where c is specific heat capacity of water. If the specific heat capacity of water (c) is $4.2 \frac{J}{g \cdot ^\circ C}$, about how much energy, in joules, is required to raise the temperature of 2.0 grams of water from $18^\circ C$ to $21^\circ C$?

- A) 16.8
- B) 12.6
- C) 25.2
- D) 33.6

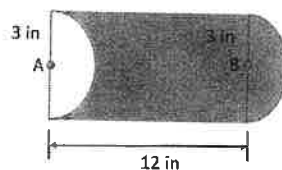
17

$$\left(-x - \frac{1}{3}\right)(mx + 1) - 3x^2 + 1$$

In the equation above, m is a constant. If the expression is equal to k , where k is a constant, what is the value of $k - m$?

- A) $\frac{7}{3}$
- B) $\frac{11}{3}$
- C) $-\frac{7}{3}$
- D) $-\frac{11}{3}$

18



In the figure above, two semi-circles, A and B, with 3 inches radii and a rectangle with 12 inches in length are shown. What is the area of the shaded region, in square inches?

- A) $36 - 9\pi$
- B) $72 - 9\pi$
- C) 36
- D) 72

19

$$\left(\frac{1}{x} + \frac{1}{y}\right)^2$$

For $x, y \neq 0$, if $x^2 + y^2 = 70$ and $xy = 6$, what is the value of the expression shown above?

- E) $\frac{41}{18}$
- F) $\frac{19}{9}$
- G) $\frac{18}{41}$
- H) $\frac{9}{19}$

20

$$m = \frac{a+b}{2}$$

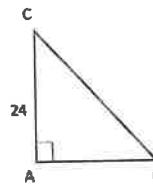
$$n = \frac{a+c}{2}$$

$$k = \frac{b+c}{2}$$

In the equations above, a , b , and c are positive constants. If $m > n > k$, which of the following inequalities is true?

- A) $a > c > b$
- B) $c > b > a$
- C) $a > b > c$
- D) $b > a > c$

22



In the right triangle ABC above, if $\cos \angle B = 0.5$, what is the value of $\tan \angle C$?

- A) $\frac{\sqrt{3}}{3}$
- B) $\frac{\sqrt{3}}{24}$
- C) $\frac{1}{2}$
- D) $\sqrt{3}$

21

$$f(x) = -\frac{x^2}{3} + 2x - l$$

In the quadratic equation above, where l is a constant. What is the value of c if $f(c) = f(5)$?

STOP

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Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

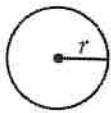
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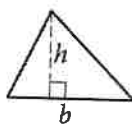


$$A = \pi r^2$$

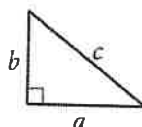
$$C = 2\pi r$$



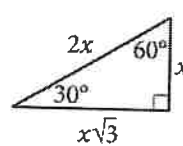
$$A = \ell w$$



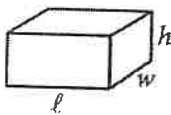
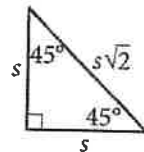
$$A = \frac{1}{2}bh$$



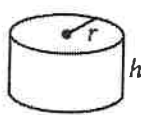
$$c^2 = a^2 + b^2$$



Special Right Triangles



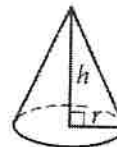
$$V = \ell wh$$



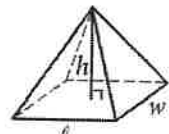
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

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1

$x - a$ is one of the factors of $3x^3 + 15x^2 - 42x$, where a is a positive constant. What is the value of a ?

- A) 2
- B) 3
- C) 7
- D) 42

2

Fitness class size	Frequency
10	20
12	18
15	10
20	7
22	5

The table above shows the distribution of 60 fitness classes size in a certain metropolitan area. What is the positive difference between the median and the mean of the data?

- A) 3.6
- B) 2.6
- C) 1.6
- D) 1.4

3

$$KE = \frac{1}{2}I\omega^2$$

The equation above shows the formula for the rotational kinetic energy (KE), where I is a rotational inertia and ω is an angular speed. If I is quadrupled and ω is halved, how does the change affect to the rotational kinetic energy, KE ?

- I) No effect on KE .
- J) KE will be doubled.
- K) KE will be $\frac{1}{2}$ times.
- L) KE will be 4 times.

4

Demographic scientist observed and interviewed the visitors to a certain shopping mall to study the characteristics of customers in a holiday. $\frac{6}{13}$ are females, $\frac{1}{5}$ are children, and $\frac{2}{7}$ are from other cities. Based on the result of survey, what number could be the total number of visitors to the shopping mall on that day?

- A) 450
- B) 453
- C) 455
- D) 458

5

Michelle hosts a party for graduation. She found out that one 2l bottle of soda can serve 8 people, one large bag of chips can serve 4 people, and one box of assorted bread can serve 12 people. If total sum of the number of sodas, large bag of chips, and box of assorted bread is 77, how many people will be at her graduation party?

- A) 160
- B) 162
- C) 165
- D) 168

6

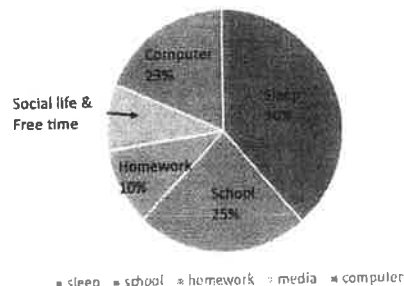
$$\left(\frac{1}{a}\right)^{x^2+2x-1} = (a^x)^{-2}$$

In the equation above, what is the absolute value of the difference of two x values, where a is a positive constant?

- M) -2
- N) 2
- O) 0
- P) 1

7

Time (24-hours) distribution of the average high school student



The circle graph shows how the average high school student spends their time 24-hours a day. Based on the graph, how many hours does the average high school student spend for social life and free time?

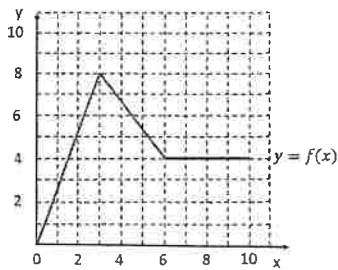
- A) 2.22
- B) 2.88
- C) 3.02
- D) 3.88

8

Julia is standing near the light post. She is 182cm tall and she is 4m away from the light post. If the height of the light post is 6m, what is the length of the shadow, in meters, of Julia? (1m = 100cm)

- A) 174.16
- B) 574.16
- C) 1.74
- D) 5.74

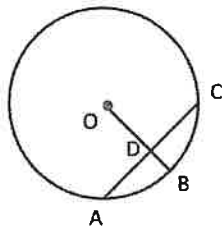
9



The figure above shows the complete graph of $y = f(x)$ in the XY -plane. The function h (not shown) is defined by $h(x) = f(x) - 3$. What is the maximum value of the function h ?

- A) 4
- B) 5
- C) 6
- D) 8

10

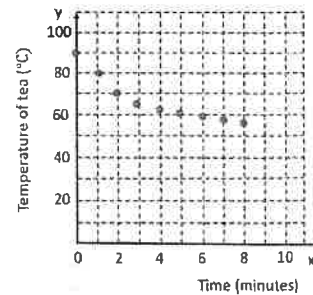


In the circle O above, the radius of a circle O is 8. If the length of BD is 2, what is the length of AC ?

- A) $2\sqrt{7}$
- B) $4\sqrt{7}$
- C) 20
- D) 10

11

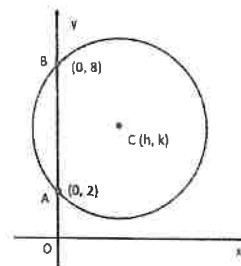
Temperature of a cup of tea vs time in minutes



In the dot plots above, it shows the temperature profile of a cup of tea over time after the cup is removed from a heat source and then left it in a room that is kept at a constant temperature. Which of the following best approximate the temperature, in $^{\circ}\text{C}$, when it is just removed from the heat source?

- A) 60
- B) 70
- C) 80
- D) 90

12



In the circle graph above, the center of the circle is $C(h, k)$ and the radius of the circle is 5, what is the value of h ?

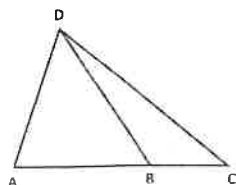
- A) 3
- B) 4
- C) 5
- D) 6

13

Gender	Short track distance			
	500m	1,000m	1,500m	3,000m
Male	23	11	9	20
Female	18	x	7	15

The table above shows the distribution of the number of Olympic short track athletes by various distances and gender in a certain country. If one of the female athletes was chosen at random, the probability that the athlete runs for 1,000m is $\frac{1}{5}$, what is the value of x ?

14



In the figure above, $\overline{AB} = 2.5\overline{BC}$. If the area of triangle ABD is 30, what is the area of triangle BCD?

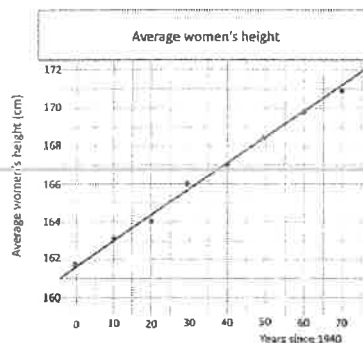
- A) 10
- B) 11
- C) 12
- D) 14

15

$$\begin{aligned} x &= 3 \\ |y - 2| &= x \end{aligned}$$

In the system of equations above, what is the sum of y values of the solutions to the system?

16



The scatter plot shows the average women's height, in centimeters, in a certain country since 1940s. A line of best fit is shown on the graph and its equation is $y = 0.133x + 161.67$, where x is the number of years after 1940. what does the line of best fit predict about the increase over the 70-year period?

- A) Every 10 years between 1940 and 2010, the average increase in women's height in that specific country is 0.133cm.
- B) For 10 years between 1940 and 2010, the total increase in women's height in that specific country is 161.67cm.
- C) Every year between 1940 and 2010, the average increase in women's height in that specific country is 0.133cm.
- D) Every year between 1940 and 2010, the average increase in women's height all over the world is 0.133cm.

17

If $f(x) = \frac{3}{4}x^2 + 1$ and $f(x - a) = \frac{3}{4}x^2 + 3x + 4$,
what is the value of a ?

- A) -2
- B) 2
- C) -1
- D) 1

19

$$\frac{1}{3x-4} + 2$$

Which of the following is equivalent to the
expression above, where $x > 2$?

- A) $\frac{6x-8}{3x-4}$
- B) $\frac{3}{3x-4}$
- C) $\frac{3x-4}{6x-7}$
- D) $\frac{3x-4}{6x-6}$

18

An oceanologist detected a sunken ship 4.5 miles
below the ocean's surface by radar. Approximately,
how many kilometers below the ocean's surface was
the sunken ship located? (1 kilometer = 0.6214
miles)

- A) 2.80
- B) 6.24
- C) 7.24
- D) 8.24

20

$$(4x^2 - ax + 1)(bx - 3)$$

The expression above is equivalent to $28x^3 - 19x^2 + 10x - 3$ for all x , where a and b are
constants. What is the value of $a + b$?

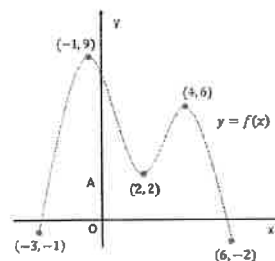
- A) 7
- B) 8
- C) 9
- D) 10

21

A math teacher scored 55 students' math exam. The teacher calculated class mean, median, range, and standard deviation of 55 scores. Later he found out that the highest score was mistakenly 2 points lower than it should have been. Which of the following remains unchanged if the teacher uses the corrected score?

- A) Mean
- B) Median
- C) Range
- D) Standard deviation

22



The complete function f is graphed in the XY -plane above. There is another function, $y = a$, where a is a constant (not shown). Which of the following could be the value of a if the system of two functions has three real solutions?

- A) 1
- B) 3
- C) 5
- D) 6

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Math

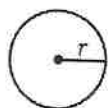
22 QUESTIONS
(TIME: 35 MIN)**DIRECTIONS**

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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- All figures lie in a plane.
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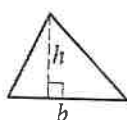
REFERENCE

$$A = \pi r^2$$

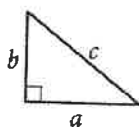
$$C = 2\pi r$$



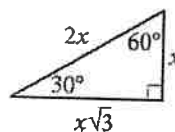
$$A = \ell w$$



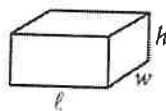
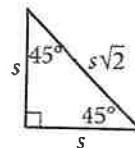
$$A = \frac{1}{2}bh$$



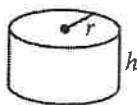
$$c^2 = a^2 + b^2$$



Special Right Triangles



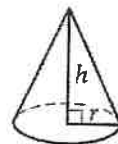
$$V = \ell wh$$



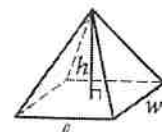
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

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- If you find **more than one correct answer**, write and circle only one answer.
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- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

$$144 = 8x + y$$

The equation above can be used to model the amount of money (y), in dollars, left after buying boba drinks for x times in a month. If $(18, 0)$ is a solution to the equation, what does it mean in this context?

- A) The unit price of a boba drink is \$18.
- B) It will take 18 days to use all the money in the budget.
- C) The person bought 18 boba drinks and no money left in the budget.
- D) The person's budget was \$18.

2

If $\frac{\sqrt{2x}}{4} = 2\sqrt{2}$, what is the value of x ?

- A) 64
- B) 16
- C) 8
- D) 4

3

Which of the following must be true if $xy < 0$?

- I. $xy \neq 0$
- II. $x^2 - y^2 < 0$
- III. $x^2 + y^2 > 0$

- A) I only
- B) I and II only
- C) I and III only
- D) III only

4

Which of the following is an example of a function whose graph has exactly one x intercept in the XY -plane?

- A) A linear function whose rate of change is zero.
- B) A quadratic function whose vertex is located at $(-1, -3)$ and opens upwards.
- C) A quadratic function whose vertex is located at the origin and opens downwards.
- D) A quadratic function whose vertex is located at $(1, 3)$ and opens upwards.

5

On a certain test, all correct answers earn the same points and all incorrect answers lose the same points, and no response earns one point. The equation $P = 5c - 2i + s$ represents a score obtained by the number of correct answers, c , and the number of incorrect answers, i , and the number of unanswered, s . If a student got 25 correct answers and 5 incorrect answers, and got 120 points on the final score, how many problems did the student leave no answers on the test?

6

$$ax^2 - 10x + 5 = 0$$

In the quadratic equation above, where a is a constant. If the equation has exactly one solution, what is the value of a ?

7

Adrian has two part-time jobs. He works as a caterer at \$25 per hour, and he also works as a math tutor at \$45 per hour. He can work no more than 30 hours per week, but he will need to earn at least \$1,500 in order to pay his rent. Which of the following inequalities represents this situation if x represents the number of hours he worked as a tutor and y represents the number of hours he worked as a caterer?

- A) $\begin{cases} x + y \leq 30 \\ 25x + 45y \leq 1,500 \end{cases}$
B) $\begin{cases} x + y \geq 30 \\ 25x + 45y \geq 1,500 \end{cases}$
C) $\begin{cases} x + y \leq 30 \\ 45x + 25y \geq 1,500 \end{cases}$
D) $\begin{cases} x + y \geq 30 \\ 45x + 25y \geq 1,500 \end{cases}$

8

In a certain college, 40% of students take fine arts classes. Of those who doesn't take fine arts classes 50% take graphic design classes. If all students in the college must take only one art related class and there are 320 students in total, how many students will take other art related classes than fine art classes or graphic design classes?

- A) 128
B) 96
C) 64
D) 48

9

$$\begin{aligned}ax - by &= 12 \\ 2x + 7y &= -36\end{aligned}$$

In the system of equations above, a and b are constants. If the system has infinitely many solutions, what is the value of $\frac{b}{a}$?

- A) $-\frac{7}{2}$
- B) $\frac{7}{2}$
- C) $-\frac{2}{7}$
- D) $\frac{2}{7}$

10

$$x^2 + 10x + y^2 - 8y + 40 = 0$$

In the circle equation above, if the circle is graphed in the XY -plane, how many units horizontally need to be translated in order to be tangent to the y -axis?

- A) 5 units to the right.
- B) 4 units to the right.
- C) 5 units to the left.
- D) 5 units to the right.

11

$$f(x) = a \cdot 2^{x-1}$$

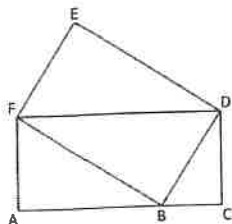
In the exponential function f above, where a is a constant. What is the value of $f(2)$ if $f(1) = 3$?

12

The height above the ground, in feet, of the riding cart in a certain roller coaster is given by $h(t) = -1.75(t - 8)^2 + 120$, t seconds after the ride starts, where $0 \leq t \leq 16$. If the riding cart is 113 ft above the ground, after how many seconds will the riding cart be at the same height, in feet, when it moves downward?

- A) 8
- B) 6
- C) 4
- D) 2

13



In the figure above, rectangle $ACDF$ is overlapped with parallelogram $BDEF$ as shown. If the area of parallelogram $BDEF$ is 50, what is the area of rectangle $ACDF$?

15

The mean of 7 numbers in a list is x . When an additional number is added to the list, the mean of all numbers is $2x - 1$. What is the value, in terms of x , of the number added to the list?

- A) $9x - 1$
- B) $8x - 8$
- C) $9x - 8$
- D) $8x - 1$

14

The landlord in a commercial building sent letters to all tenants that the lease amount will be raised 12% every year. If a tenant is paying \$15,000 for one-year lease in the beginning of 2020, Approximately, how much will the tenant need to pay in the beginning of 2023?

- A) \$18,816
- B) \$20,400
- C) \$21,074
- D) \$23,603

16

If $x^2 + y^2 = a$ and $xy = b$, which of the following is equivalent to $-4b + 2a$?

- A) $-2(y - x)^2$
- B) $2(y - x)^2$
- C) $4(x - y)^2$
- D) $-4(y - x)^2$

17

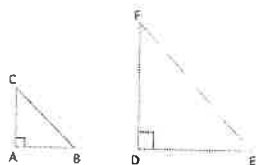
The amount of Carbon in milligrams.					
	A	B	C	D	E
Basalt	0.41	0.40	0.42	0.39	0.41
Sandstone	0.02	0.54	0.35	0.42	0.13

Five samples of each of two different rock types are collected from a mountain in California. The values in the table shows the amount of Carbon, in milligrams, found in 1 gram of each rock sample. Which of the following statements is valid based on the values in the table?

- I. The standard deviation of Basalt samples is greater than that of Sandstone samples.
- II. The median of Sandstone samples is smaller than that of Basalt samples.
- III. The range of Sandstone samples is greater than that of Basalt samples.

- A) I only
- B) I and II only
- C) II and III only
- D) III only

18



In the right triangles above, if two triangles are similar and $AB = 3$ and $BC = 5$, then what is the value of $\sin \angle E$?

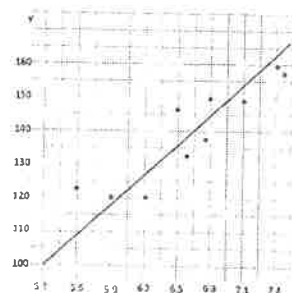
- A) $\frac{3}{4}$
- B) $\frac{2}{5}$
- C) $\frac{3}{5}$
- D) $\frac{4}{5}$

19

A right triangular prism has a height of 8 cm and the area of right triangle (Base) is 10 cm^2 . If the mass of the prism is 146 g, what is the density, in g/cm^3 , of the prism?

- A) 1.825
- B) 3.650
- C) 0.548
- D) 0.274

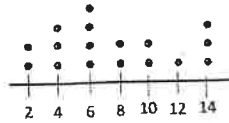
20



The scatter plot above shows the distribution of heights (y values), in centimeters, of women based on their shoe sizes (x values). The line of best fit is also drawn in the XY-plane. What is the number of women for which the line of best fit predicts a value greater than the actual value?

21

Data Set A



Data set A is shown above. Data set B is created by adding 2 to each of the values in the data set A. Which of the following statements is valid for the data?

- A) The range of data set A is smaller than that of data set B.
- B) The mean of data set A is equal to that of data set B.
- C) The median of data set A is equal to that of data set B.
- D) The standard deviation of data set A is equal to that of data set B.

22

A survey was conducted randomly at a local shopping mall to determine the proportion of people who know a recent new dancing boys' group. Out of 360 participants sampled, 234 knew the dancing boys' group. The margin of error of the survey was 5%. Which of the following statements best represents a possible conclusion about the percentage of people who know the dancing boys' group?

- A) Exactly 65% of the people in the mall know the dancing boys' group.
- B) More than 70% of the people in the mall know the dancing boys' group.
- C) Less than 60% of the people in the mall don't know the dancing boys' group.
- D) It is plausible that between 60% and 70% of the people in the mall know the dancing boys' group.

STOP

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Math

22 QUESTIONS

(TIME: 35 MIN)

DIRECTIONS

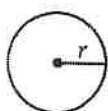
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REFERENCE

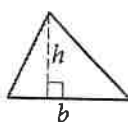


$$A = \pi r^2$$

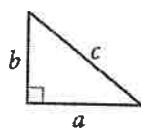
$$C = 2\pi r$$



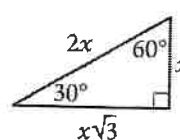
$$A = \ell w$$



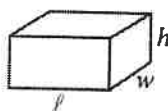
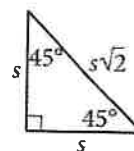
$$A = \frac{1}{2}bh$$



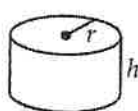
$$c^2 = a^2 + b^2$$



Special Right Triangles



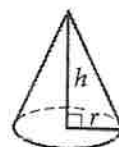
$$V = \ell wh$$



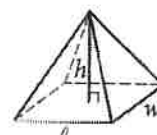
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

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1

In the xy -plane, what is a point of intersection between two graphs of $y = 2x - 4$ and $y = x^2 + x - 6$?

- A) $(-2, -8)$
- B) $(2, 0)$
- C) $(1, -2)$
- D) $(-1, 2)$

2

Function f is graphed in the xy -plane by translating the function h up 4 units. If the function h is defined by $h(x) = 3x^2 - x$, what is the value of $f(2)$?

- A) 10
- B) 12
- C) 14
- D) 6

3

There are 620 students in ABC college. 110 of the students were randomly chosen and surveyed on whether or not they will travel abroad over the summer. The results of the survey show that 66 of the surveyed students were planning to travel abroad over the summer. The margin of error on this survey is $\pm 5\%$. What is the range of students in the entire college that would travel abroad over the summer?

- A) 55-65
- B) 79-141
- C) 341-403
- D) 217-279

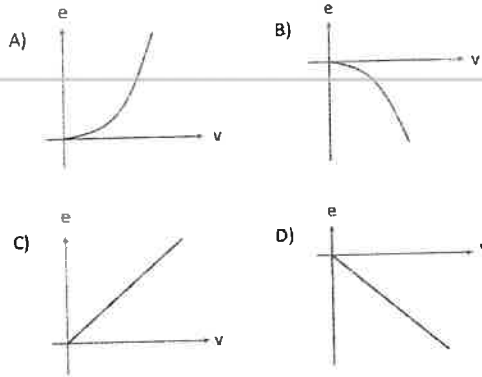
4

A technician manufactured 6 more lamps on Tuesday than he manufactured on Monday. If he manufactured 20% more lamps on Tuesday than on Monday, how many lamps did he manufacture on Tuesday?

5

Kinetic Energy (J)	Velocity (m/s)
100	10
400	20
900	30
1,600	40

In physics lab, a student found that the velocity of a miniature cart affected the kinetic energy, as shown in the table above. Which of the following graphs best represents the relationship between the velocity (v) of cart, in m/s, and the kinetic energy (e), joule, as indicated in the table?



6

$$\frac{3x-3}{(x-3)^2} - \frac{3}{x-3}$$

If the expression above is equivalent to $\frac{k}{(x-3)^2}$, where k is a constant and $x \neq 2$, what is the value of k ?

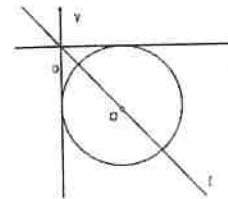
7

Ages	Frequency
17	50
18	150
19	170
20	80
21	34
22	16

The table above shows the distribution of ages of 500 freshmen in a certain college. What is the positive difference between the mean and the median of the data in the table?

- A) 0.500
- B) 0.108
- C) 1.080
- D) 5.000

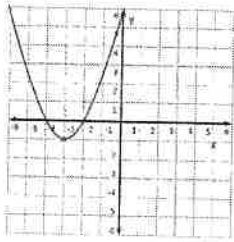
8



A circle O with radius 3 is tangent to both x -axis and y -axis in the xy -plane as shown above. If a line l passes through the origin and the center of the circle O , what is the equation of the line l ?

- A) $y = x + 3\sqrt{2}$
- B) $y = x$
- C) $y = -x - 3\sqrt{2}$
- D) $y = -x$

9



The graph of quadratic function of $y = f(x)$ is shown in the xy -plane above. Which of the following equations shows the minimum value of the function in the equation?

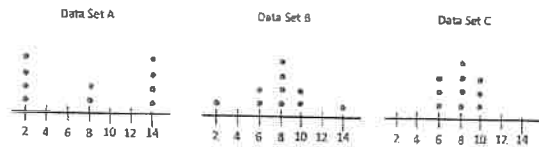
- A) $y = (x + 2)(x + 4)$
- B) $y = x^2 + 6x + 8$
- C) $y = x(x + 6) + 8$
- D) $y = (x + 3)^2 - 1$

10

Veronica has 1.2 miles left after running k percent of the entire trail in the morning. Which of the following expressions represents the distance of the entire trail in terms of k ?

- A) $\frac{120}{100-k}$
- B) $\frac{120}{100-k}$
- C) $\frac{120-1.2k}{100}$
- D) $\frac{100}{120-k}$

11



The dot plots of three data sets are shown above. Which data set has the least standard deviation?

- A) Data set A
- B) Data set B
- C) Data set C
- D) All data sets have the same standard deviation.

12

If $f(x - 1) = x^2 - 3x - 2$ for all values of x , which of the following could be the function f ?

- A) $f(x) = x^2 - 5x + 2$
- B) $f(x) = x^2 - 3x - 3$
- C) $f(x) = x^2 - x - 4$
- D) $f(x) = x^2 + x - 4$

13

The census shows that the number of labor force population in a certain country-side town is reduced by one-fifth every year. If the number of labor force population in the country-side town t years after the first census surveyed is represented by P and the initial labor population on the country-side town was k , where k is a constant, which of the following equations best represents this situation after t years of the first census surveyed?

- A) $P(t) = \frac{1}{5}k^t$
- B) $P(t) = k\left(\frac{1}{5}\right)^t$
- C) $P(t) = t\left(\frac{1}{5}\right)^k$
- D) $P(t) = k\left(\frac{4}{5}\right)^t$

14

$$f(x) = 2\left(\frac{1}{4}\right)^x - 2$$

If the exponential function of $y = f(x)$ is graphed in the xy -plane, what is the coordinates of x -intercept of the function?

- A) $(0, 0)$
- B) $(1, 0)$
- C) $(-1, 0)$
- D) $(-2, 0)$

15

Two triangles, $\triangle ABC$ and $\triangle KML$, are similar. The area of triangle $\triangle ABC$ is 46 in^2 . If the length of sides $AB = 4 \text{ in}$ and $KM = 8 \text{ in}$, what is the area of $\triangle KML$?

- A) 184
- B) 92
- C) 46
- D) 23

16

Circle A: $(x - 1)^2 + y^2 = 4$

Circle B: $x^2 - 2x + y^2 - c = 0$

In the xy -plane, two circles A and B are co-centered. If the radius of circle A is two less than the radius of circle B, what is the value of constant c in the equation of circle B?

- A) 36
- B) 35
- C) 16
- D) 15

17

A triangle has a height that is 125% of its base. If the height is increased by 2 cm, how much does the base need to change in order to keep the same proportion of the original triangle?

- A) The base needs to increase 1.6 cm.
- B) The base needs to decrease 1.6 cm.
- C) The base needs to increase 2 cm.
- D) The base needs to decrease 2 cm.

18

A quadratic function f is graphed in the xy -plane. If one of the x -intercept is $(2, 0)$ and the vertex is located at $(4, 3)$, what is the coordinates of the other x -intercept of the function?

- A) $(4, 0)$
- B) $(5, 0)$
- C) $(6, 0)$
- D) $(7, 0)$

19

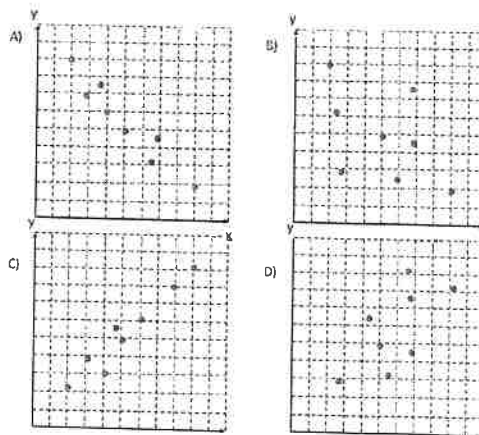
If $x > 0$, which of the following is equivalent to $\sqrt[7]{x^5}$?

- I. $x^{\frac{5}{7}}$
- II. $\frac{x}{x^{\frac{2}{7}}}$
- III. $x \cdot (x^{\frac{1}{7}})^{-2}$

- A) I only
- B) I and II only
- C) I and III only
- D) I, II, and III

20

Which of the following scatter plots best represent a strong negative association between x and y ?



21

For $x > 0$ and $y > 0$, the quotient of the square root of x and twice the square of y is 56, and the cube root of the sum of 5 and y is the square of x . Which of the following equations best represents the statements above.

- A) $\begin{cases} \frac{\sqrt{x}}{2y^2} = 56 \\ (5 + y)^3 = \sqrt{x} \end{cases}$
- B) $\begin{cases} \frac{\sqrt{x}}{(2y)^2} = 56 \\ \sqrt[3]{(5 + y)} = \sqrt{x} \end{cases}$
- C) $\begin{cases} \sqrt[3]{(y + 5)} = \sqrt{x} \\ \frac{\sqrt{x}}{2y^2} = 56 \end{cases}$
- D) $\begin{cases} \frac{\sqrt{x}}{2y^2} = 56 \\ \sqrt[3]{(y + 5)} = x^2 \end{cases}$

22

A right triangle has angles measuring 45° , 45° , and 90° . If the length of the hypotenuse is 4, what is the area of the right triangle?

STOP

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Math

22 QUESTIONS

(TIME: 35 MIN)

DIRECTIONS

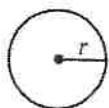
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NOTES

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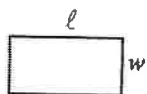
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REFERENCE

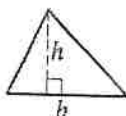


$$A = \pi r^2$$

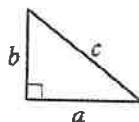
$$C = 2\pi r$$



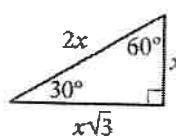
$$A = \ell w$$



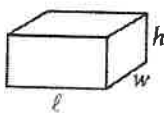
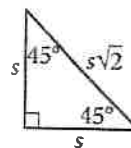
$$A = \frac{1}{2}bh$$



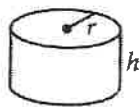
$$c^2 = a^2 + b^2$$



Special Right Triangles



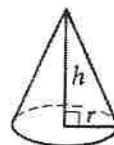
$$V = \ell wh$$



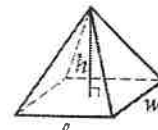
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

x	y
12	56
18	80
27	116

The table above shows some values of x and their corresponding values of a linear function, $y = f(x)$. Line l is formed by translating $y = f(x)$ up 7 units in the XY -plane. What is the y -intercept of the line l ?

- A) (15,0)
- B) (0,15)
- C) (8,0)
- D) (0,8)

2

$$\begin{aligned} -x + 3y &= 2 \\ 8x - 24y &= -16 \end{aligned}$$

In the system of equations above, for any real number k , which of the following set of points lies on the graph of the system in the XY -plane?

- A) $\left(k, \frac{2-k}{3}\right)$
- B) $\left(k, \frac{k+2}{3}\right)$
- C) $(3k+2, k)$
- D) $(3k-1, k)$

3

The area of an equilateral triangle is $\sqrt{3} \text{ cm}^2$. The height of the triangle is $k\sqrt{3} \text{ cm}$, where k is a constant. What is the value of k ?

- A) 3
- B) 2
- C) 1
- D) $\frac{1}{2}$

4

$$p = 20 + 2x$$

The equation above gives the speed (p), in meters per second, of water x meters down from the top of the mountain. Which of the following best represents 20 in this context?

- A) The initial speed of water, in meters per second, at the top of the mountain.
- B) The initial height of mountain, in meters, at the top of the mountain.
- C) Average increase of the speed, meters per second, of water per second at the top of the mountain.
- D) Average speed of water, in meters per second, for the entire trip from the top of the mountain to the bottom.

5

The function f is defined by $f(x) = -\frac{1}{7}x^2$. In the XY -plane, the graph of $y = g(x)$ is the result of translating $f(x)$ 3 units down. Which of the following shows the equation of $g(x)$?

- A) $g(x) = -\frac{1}{7}x^2 + 3$
- B) $g(x) = -\frac{1}{7}(x - 3)^2$
- C) $g(x) = -\frac{1}{7}x^2 - 3$
- D) $g(x) = -\frac{1}{7}(x + 3)^2$

6

$$3(x + k) = ax + b$$

In the equation above, a , b , and k are constants. If the equation has infinitely many solutions, which of the following must be equal to b ?

- A) k
- B) $3k$
- C) 3
- D) $3a$

7

$$P(t) = 15(1.2)^t$$

The equation above gives the estimated number of customers at a restaurant, where t is the number of years since the restaurant opened. Which of the following best interprets the number 1.2 in the context?

- A) The estimated number of customers is increased by 1.2% every year.
- B) The estimated number of customers is increased by 1.2 every year.
- C) The estimated number of customers is increased by 20% every year.
- D) The estimated number of customers is decreased by 20% every year.

8

If $4x - 6y = 14t$, which of the following expressions is equivalent to $4x^2 - 12xy + 9y^2$?

- A) $49t^2$
- B) $7t$
- C) $7t^2$
- D) $49t$

9

$$f(x) = -3(x - 1)^2 + 5$$

In the function above, what is the maximum value of the function?

- A) 1
- B) 2
- C) 5
- D) -3

10

The equation $h = 24.5 - 2.1t$ approximate the height h , in yard, of a ball in free fall t seconds after it is dropped from the height of 24.5 yard from the ground. What is the decrease in height, in yard, 1 second after it was dropped?

- A) $\frac{1}{2.4}$
- B) 2.4
- C) 22.4
- D) 2.1

11

A wedding planner is planning a party dinner for guests. It costs the wedding planner a onetime fee of \$500 to rent the hall and \$25.75 per attendee. If the wedding planner has a budget of \$5,000. What is the greatest number of attendees for a party dinner within the budget?

- A) 176
- B) 175
- C) 174
- D) 173

12

A cylinder is inscribed in a cube such that the height of cylinder is equal to the length of side of the cube and the circumference of circular base of cylinder touches four sides of square base of the cube. If the side of the cube is a , what is the volume of the space in the cube not taken by the cylinder?

- A) $a^3 \left(1 - \frac{\pi}{2}\right)$
- B) $a^3 \left(1 - \frac{\pi^2}{4}\right)$
- C) $a^3 \left(1 - \frac{\pi}{4}\right)$
- D) $a \left(a^2 - \frac{\pi}{2}\right)$

13

Which of the following is equivalent to $x^{\frac{5}{7}}$, where $x > 0$?

- A) $\sqrt[56]{x^{40}}$
- B) $\sqrt[12]{x^{40}}$
- C) $\sqrt[40]{x^{56}}$
- D) $\sqrt[40]{x^{12}}$

14

The amount of money Tom has is 6% less than the amount of money Jerry has. If the amount of money Tom has is k times the amount of money Jerry has, what is the value of k ?

- A) 0.06
- B) 6
- C) 94
- D) 0.94

15

The value of an oil painting drawn by a famous artist increased 1% of its value of the previous year annually. Which of the following best models regarding the value of the oil painting over time?

- A) Linear decreasing
- B) Exponential increasing
- C) Linear increasing
- D) Exponential decreasing

16

$$2x^2 - 4x + 2y^2 + 8y = -4$$

The circle equation above is graphed in the XY -plane. The circle passes through two points on the y -axis. What is the sum of the y coordinates of two points?

- A) -4
- B) 4
- C) $2\sqrt{2}$
- D) $-2\sqrt{2}$

17

$$\begin{aligned}4x - 7y &= 2 \\ kx + my &= 5\end{aligned}$$

In the system of equations, k and m are constants. If two linear graphs are perpendicular in the XY -plane, which of the following system of equations also represents two perpendicular linear graphs in the XY -plane?

- A) $\begin{cases} 8x - 7y = 2 \\ kx + 2my = 6 \end{cases}$
- B) $\begin{cases} 8x - 7y = 9 \\ kx + my = 6 \end{cases}$
- C) $\begin{cases} 4x - 7y = 5 \\ 2kx + my = 0 \end{cases}$
- D) $\begin{cases} 4x - 7y = 4 \\ kx - 2my = 1 \end{cases}$

18

If the length of a diagonal of a square is $3\sqrt{2}$ inch, what is the perimeter of the square, in inches?

- A) $12\sqrt{2}$
- B) 12
- C) 24
- D) $24\sqrt{2}$

19

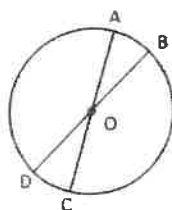
A soccer ball is kicked from the top of a building. The equation $h(t) = -4.9t^2 + 6t + 12$ represents the height, in meters, of the ball from the ground t seconds after it is kicked. What does the number 12 represent in the equation?

- A) The height, in meters, of the building from the ground to the top of the building.
- B) The distance the soccer ball can travel until it hit the ground.
- C) The maximum height of the soccer ball can reach after it was kicked.
- D) The time, in seconds, the soccer ball to reach the ground after it was kicked.

20

A school teacher has a budget \$700 to purchase wooden pencils and mechanical pencils. The teacher must purchase a minimum of 500 pencils to give out to all students in class. If the cost of a wooden pencil is \$0.70 and the cost of a mechanical pencil is \$2.50, what is the maximum number of mechanical pencils the teacher can purchase to stay within the budget?

21



In the circle O above, the diameter of the circle is 10. If $\angle BOC = 5\angle AOB$, what is the length of an arc $\widehat{CD}$?

- A) $\frac{1}{6}\pi$
- B) $\frac{25}{12}\pi$
- C) $\frac{5}{6}\pi$
- D) $\frac{1}{2}\pi$

22

An electric car manufacturing company produced their electric cars 200% more than its number of electric cars produced in the previous year from 2,000 to 2,020. A model formulated based on the information above to compute the number of electric cars, $p(x)$, at the end of x years after the company first started to produce in the year 2,000, where $0 \leq x \leq 20$. If $p(0) = 130$, Which of the following functions best models this situation?

- A) $p(x) = 130(2)^x$
- B) $p(x) = 260(2)^x$
- C) $p(x) = 130(3)^{x-1}$
- D) $p(x) = 130(3)^x$

STOP

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Math

22 QUESTIONS
(TIME: 35 MIN)

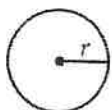
DIRECTIONS

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NOTES

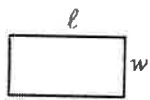
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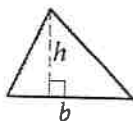
REFERENCE

$$A = \pi r^2$$

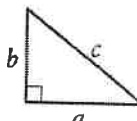
$$C = 2\pi r$$



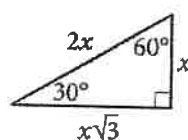
$$A = \ell w$$



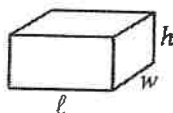
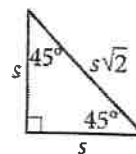
$$A = \frac{1}{2}bh$$



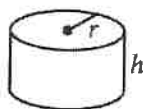
$$c^2 = a^2 + b^2$$



Special Right Triangles



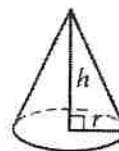
$$V = \ell wh$$



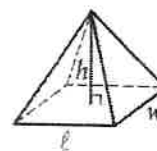
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

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The sum of the measures in degrees of the angles of a triangle is 180.

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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

$$\begin{aligned} AB &= 25 \\ BC &= 60 \\ AC &= 65 \end{aligned}$$

In a triangle ABC, the lengths of all sides are shown as above. If the triangle KLM is similar to the triangle ABC, where A corresponds to K and B corresponds to L, what is the value of $\sin K$?

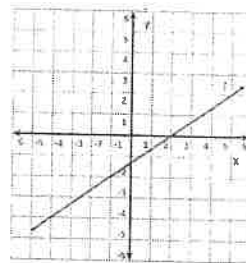
- A) $\frac{5}{12}$
- B) $\frac{12}{13}$
- C) $\frac{5}{13}$
- D) $\frac{12}{13}$

2

The number k is 80% less than the number m and the number m is 120% greater than 58. What is the value of k ?

- A) 55.68
- B) 13.92
- C) 25.52
- D) 102.08

3



The graph of line l is shown above. If the equation of the line l is $y = h(x) - 5$, which of the following defines $h(x)$?

- A) $h(x) = \frac{2}{3}x - \frac{4}{3}$
- B) $h(x) = \frac{2}{3}x + \frac{11}{3}$
- C) $h(x) = \frac{2}{3}x + \frac{7}{2}$
- D) $h(x) = \frac{2}{3}x - \frac{13}{2}$

4

One gallon of gasoline will cover 6 day-use of a power generator. The north pole explorers need to use z days of power generator in order to complete their project. Which of the following equations enable to figure out the total amount of gasoline (g), in liters, needed to finish the project? (1 gal = 3.785 l)

- A) $g = \frac{(3.785) \cdot (z)}{6}$
- B) $g = \frac{6}{(6) \cdot (3.785)}$
- C) $g = \frac{(3.785) \cdot (6)}{z}$
- D) $g = \frac{(6) \cdot (z)}{3.785}$

5

There are 67 students in international culture club in a certain college. A sample of the international culture club were selected randomly and asked whether they intend to visit other countries over the summer break. Of those surveyed, about 30% responded that they intend to visit other countries over the summer. Based on the result of this survey, which of the following best estimates the total number of students in the international club who does NOT intend to visit other countries over the summer?

- A) 20
- B) 47
- C) 30
- D) 67

7

$$2a - \frac{1}{6b} = 1 - \frac{1}{c}$$

The given equation relates the positive real numbers a , b , and c . Which of the following equations correctly solve for c in term of a and b ?

- A) $c = \frac{1}{6b-12ab+6a}$
- B) $c = \frac{6b}{6b-12ab+1}$
- C) $c = \frac{12ab}{6b+1}$
- D) $c = \frac{6b}{12ab+6b+1}$

6

Kaitlyn participated Honolulu triathlon Olympic in Hawaii. She ran at the average speed of 4.5 miles per hour for r hours and swam at the average speed of 2 miles per hour for s hours and cycled at the average speed of 43 miles per hour for c hours for a combined total distance of 32 miles. Which of the following equations correctly represents this situation?

- A) $4.5r + 2c + 43s = 32$
- B) $\frac{4.5}{r} + \frac{2}{s} + \frac{43}{c} = 32$
- C) $4.5r + 43c + 2s = 32$
- D) $4.5r + 43c + 2s = 96$

8

$$\begin{aligned} y &\leq -2 \\ x &< 2y - 4 \end{aligned}$$

If the point $(-10, k)$ is a solution to the system of inequalities in the XY -plane. Which of the following could not be the value of k ?

- A) 0
- B) 2
- C) -2
- D) -3

9

$$f(x) = 5,000(1.13)^x$$

The given function f models the number of hand fans a company manufactured at the end of each year, where x is the number of years since the end of 2001, where $0 \leq x \leq 10$. If $y = f(x)$ is graphed in the XY -plane, what does y -intercept represent in this context?

- A) The number of hand fans the company manufactured at the end of 2001.
- B) The percent increase in the number of hand fans the company manufactured since at the end of 2001.
- C) The minimum number of hand fans the company could manufacture at the end of 2001.
- D) The maximum number of hand fans the company could manufacture at the end of 2001.

10

$$-\frac{1}{2}x^2 - 3x + 4 = 0$$

In the quadratic equation above, how many real solutions the given equation have?

- A) Zero
- B) Infinitely many
- C) One
- D) Two

11

$$y = \frac{2}{3}x - 3$$

In the system of linear equations, one of the equations is given above. If the system has no solution in the XY -plane, which of the following could be the other equation in the system?

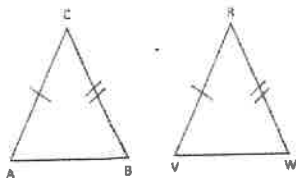
- A) $2x + 3y = 9$
- B) $2x - 3y = 0$
- C) $3x - 2y = -1$
- D) $-3x - 2y = 4$

12

Which of the following circle equations will intersect the x -axis at exactly one point?

- A) $(x - 2)^2 + (y - 1)^2 = 1$
- B) $(x - 1)^2 + (y + 1)^2 = 2$
- C) $(x + 2)^2 + y^2 = 1$
- D) $x^2 + y^2 = 1$

13



In the two triangles above, if $\overline{AC} \cong \overline{VR}$, $\overline{BC} \cong \overline{WR}$, Which of the following additional piece of information is sufficient to prove whether two triangles are congruent?

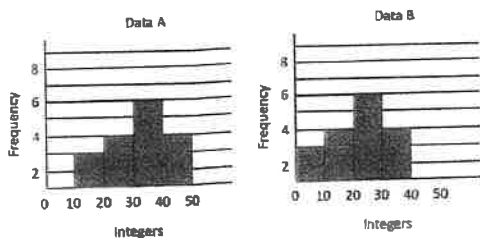
- A) $\angle A \cong \angle V$
- B) $\angle C \cong \angle R$
- C) $\angle B \cong \angle W$
- D) $\angle A \cong \angle W$

15

$$f(x) = \frac{2}{(2-x)^2 + 4(2-x) + 4}$$

In the rational expression above, what is the sum of x values which make the function f undefined?

14

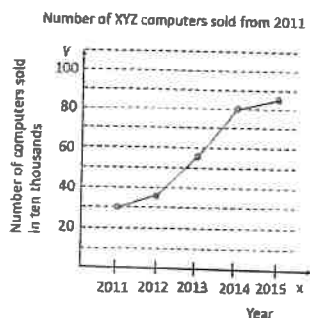


Two sets of 17 integer distribution each is shown in the histogram above. For each of the histograms, for example, the first interval in data A represents the frequency of the integers greater than or equal to 10, but less than 20. What is the greatest possible difference to the nearest whole number between the mean of data A and the mean of data B?

16

Kyle opened a bank account which earns 4% annual interest. His initial deposit amount was \$1,300. He wants to find the balance of the account after t years. If he set up the equation, $Balance = 1,300(x)^t$, what is the value of x in the equation?

17



According to the graph above, the number of computers sold in 2012 is what fraction of the number of computers sold in 2014?

- A) $\frac{3}{8}$
- B) $\frac{7}{17}$
- C) $\frac{7}{16}$
- D) $\frac{6}{17}$

18

Min paid \$200,000 in total for his sport car. He made a down payment of \$50,000 plus M monthly payments of \$ x each. Which of the following equations could represent this situation?

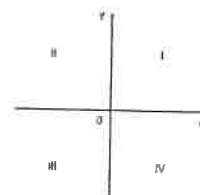
- A) $200,000 = 50,000x + M$
- B) $200,000 = 50,000M + x$
- C) $200,000 = 50,000xM$
- D) $200,000 = 50,000 + xM$

19

Mary Ellen rent a truck at a cost of \$65 per day plus one-time charge of \$45 for insurance and processing fee. Which of the following equations could represents the total cost, $f(x)$, in dollars, to rent a truck for x days?

- A) $f(x) = 65 + 45x$
- B) $f(x) = (65 + 45)x$
- C) $f(x) = 45 + 65x$
- D) $f(x) = 45 + (65 + 45)x$

20



$$y < 1$$

$$2x + y \leq -8$$

If the system of inequalities is graphed in the xy -plane above, which quadrant contains no solutions to the system?

- A) I
- B) II
- C) III
- D) IV

21

A survey was conducted by an electric car company found that the average battery life of a random sample is 9.8 years with an associated margin of error is 0.7 years. Then another survey was conducted with a much larger size of sample size, with the mean and margin of error of the new sample size being calculated in the same way as the first survey. Which of the following is valid most likely?

- A) The margin of error from the new survey would be larger than the margin of error from the first survey.
- B) The margin of error from the new survey would be smaller than the margin of error from the first survey.
- C) The mean of the new survey must be larger than the mean of the first survey.
- D) The mean of the new survey must be smaller than the mean of the first survey.

22

Square A has a perimeter of 16 inches. The length a side of square B is three times the length of a side of square A. What is the length of a diagonal of square B, in inches?

- A) $6\sqrt{2}$
- B) $12\sqrt{2}$
- C) $16\sqrt{2}$
- D) $18\sqrt{2}$

STOP

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Math

22 QUESTIONS

(TIME: 35 MIN)

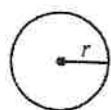
DIRECTIONS

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NOTES

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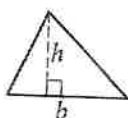
REFERENCE

$$A = \pi r^2$$

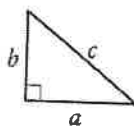
$$C = 2\pi r$$



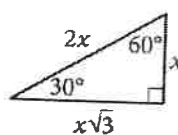
$$A = \ell w$$



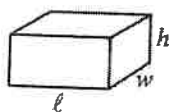
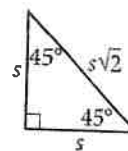
$$A = \frac{1}{2}bh$$



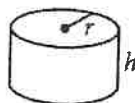
$$c^2 = a^2 + b^2$$



Special Right Triangles



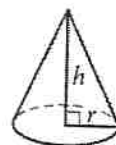
$$V = \ell wh$$



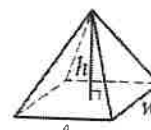
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

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- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

Suppose that a cows give b gallons of milk in k days. At this rate, how many days will take for m cows to give n gallons of milk?

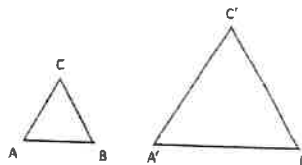
- A) $\frac{amnk}{b}$
 B) $\frac{amk}{bn}$
 C) $\frac{ank}{bm}$
 D) $\frac{bm}{ak}$

2

The skyline rider in amusement park charges \$7 for a child and \$12 for an adult for a ride. If a riding operator collected \$4,150 from 450 riders in one day, which of the following system of equations could be used to determine the number of children riders, x , and the number of adults riders, y , for the day?

- A) $\begin{cases} x + y = 450 \\ 7y + 12x = 4,150 \end{cases}$
 B) $\begin{cases} x + y = 450 \\ 12y + 7x = 4,150 \end{cases}$
 C) $\begin{cases} x + y = 4,150 \\ 7x + 12y = 450 \end{cases}$
 D) $\begin{cases} x + y = 4,150 \\ 12x + 7y = 450 \end{cases}$

3



In the two similar triangles above, if triangle ABC is dilated by a factor of 2 to form another triangle $A'B'C'$. Which of the following statements is NOT valid?

- A) The length of side $A'B'$ is twice the length of side AB .
 B) The measure of angle C is a half of the measure of angle C' .
 C) The perimeter of triangle $A'B'C'$ is twice the perimeter of triangle ABC .
 D) Angle A is congruent to angle A' .

4

In the XY -plane, the function f has two x -intercepts at $(2, 0)$ and $(6, 0)$ and one y -intercept at $(0, -4)$. Which of the statements is true for the function f ?

- I. $-f(0) = 4$
 II. $f(6) = 0$
 III. $f(0) = 2$

- A) I only
 B) II only
 C) I and II only
 D) II and III only

5

$$f(x) = x^2 - 2x + 3$$

$$g(x) = x^2$$

In two functions, f and g , above in the xy -plane, which of the following statements describes correctly the relationship between two functions?

- A) $f(x)$ is formed when $g(x)$ is translated 1 unit right and 3 units up.
- B) $f(x)$ is formed when $g(x)$ is translated 2 units right and 2 units up.
- C) $f(x)$ is formed when $g(x)$ is translated 1 unit left and 2 units up.
- D) $f(x)$ is formed when $g(x)$ is translated 1 unit right and 2 units up.

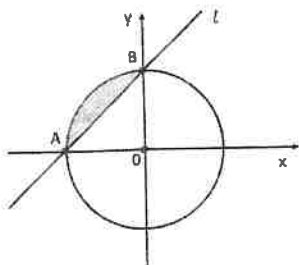
7

x	$f(x)$
-1	6
$\frac{1}{2}$	9

The table gives some values of x and the corresponding values of $f(x)$. If a linear function $f(x)$ is graphed in the XY -plane, what is the coordinates of x intercept of the function?

- A) (0, 8)
- B) (8, 0)
- C) (-4, 0)
- D) (0, -4)

6



A circle with area $25\pi \text{ in}^2$ is centered at the origin in the XY -plane shown above. A line l passes through x , y intercepts, A and B , respectively. What is the area, in square inches, of shaded region?

- A) $\frac{25(\pi-1)}{4}$
- B) $\frac{25(\pi-2)}{4}$
- C) $\frac{25(\pi-2)}{2}$
- D) $\frac{25(\pi-50)}{4}$

8

$$\frac{5x^2 - 16}{x^2 - 4} - \frac{8}{x + 2} = \frac{1}{x - 2}$$

In the equation above, what is the value of x ?

- A) 2
- B) $\frac{1}{5}$
- C) $-\frac{1}{5}$
- D) -2

9

$$\frac{m^{\frac{2}{3}} \cdot (m^{-2})^{\frac{1}{4}}}{(m \cdot m^{\frac{1}{2}})^{-2}} = m^n$$

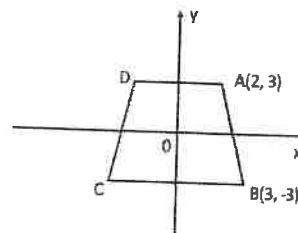
In the equation above, where m and n are positive constants. Which of the following is the value of n ?

- A) $\frac{19}{6}$
- B) $\frac{1}{6}$
- C) $\frac{13}{6}$
- D) $\frac{7}{6}$

10

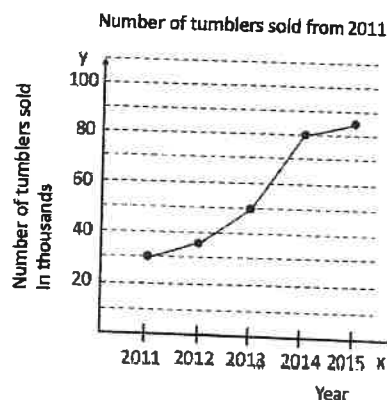
Elliott plans to rent a scooter near ocean. The scooter's hourly rental cost is \$65. And he will also need to pay \$25 for one-time cleaning & set fee. If he has a budget of \$300 to spend and the rental is available only for a whole number of hours, what is the maximum number of hours that he can rent a scooter?

11



An isosceles trapezoid ABCD is graphed in the XY-plane above. If two points, A and D, are symmetrical with respect to y axis, what is the slope of the segment $\overline{CD}$?

12



According to the line graph above, between which two consecutive years was the greatest rate of change in the number of tumblers sold?

- A) 2011-2012
- B) 2012-2013
- C) 2013-2014
- D) 2014-2015

13

The weight of an object on Mars is about 40% of the weight on Earth. The weight of an object on Mars is approximately $\frac{1}{9}$ the weight on Venus. If an object is 180 pounds on Earth, what is the weight, in pounds, of the object on Venus?

- A) 8
- B) 4050
- C) 408
- D) 648

14

The local adult school in a certain town plans to increase its language programs by x classes every six months. If there are p classes available now at this adult school, which of the following best models the total number of classes, y , the adult school will have t years from now?

- A) $y = xp + t$
- B) $y = tp + x$
- C) $y = xp + t$
- D) $y = 2xt + p$

15

Majors	Professional Activity		Total
	Research	Teaching	
Humanities	75	175	250
Technical	350	200	550
Total	375	375	800

A survey was conducted on 800 Professors' activities in a county. The results are summarized in the table above. If one of the professors is selected at random in the survey, which of the following is the probability that the professor is teaching in technical majors?

- A) 0.36
- B) 0.70
- C) 0.25
- D) 0.22

16

$$y = -x^2 + 8$$

The graph of the quadratic equation above intersects with the graph of a line l at two points $A(m, 4)$ and $B(b, -8)$ in the XY -plane. What is the greatest possible value of the slope of line l ?

17

A community social worker surveyed 200 adults selected at random from a large city and asked whether or not they are satisfied with their current marriage life. Of those surveyed, 65% responded that they were satisfied with their current marriage life. Based on the results of this survey, which of the following statements must be true?

- I. If another survey is conducted with the same sample size in a different city, 65% would respond they are satisfied with current marriage.
- II. If another survey is conducted in a different size sample size in the same city, 65% would respond they are satisfied with current marriage.
- III. If all adults in the same city are surveyed, 65% would respond they are satisfied with current marriage.

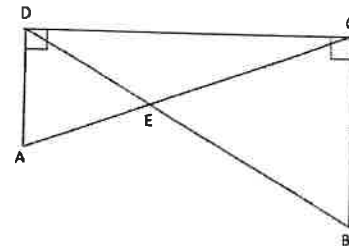
- A) I only
- B) I and II only
- C) I and III only
- D) None

18

$$f(x+1) = \frac{f(x-2) \cdot f(x-1)}{f(x)}$$

The function f has a relation above. If $f(1) = 2$, $f(2) = 3$, $f(3) = 4$, what is the value of $f(5)$?

19



In the figure above, $AD = 5$, $BC = 7$, and $AC = 13$. What is the difference between the area of triangle BCE and the area of triangle ADE?

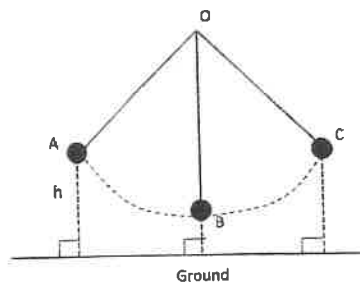
20

$$\frac{x}{x+y} = k$$

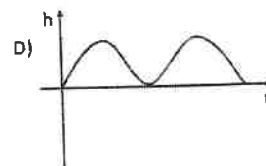
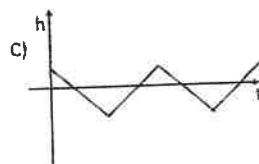
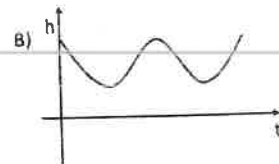
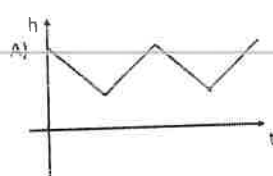
In the equation above, if x, y are negative integers, which of the following is true about k ?

- A) $k < 0$
- B) $k > 1$
- C) $0 < k < 1$
- D) $-1 < k < 0$

22



The figure above shows the position of a pendulum when it swings back and forth from A to C. Assuming no air friction, which of the following best represents the height of the pendulum if the pendulum release from the position A?



21

If the radius of the base of a cone-shaped container is increased by 100%, by what percent of the height of the container must be decreased in order to keep the same volume?

STOP

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Math

22 QUESTIONS

(TIME: 35 MIN)

DIRECTIONS

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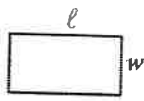
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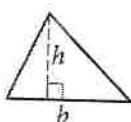


$$A = \pi r^2$$

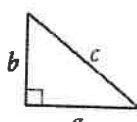
$$C = 2\pi r$$



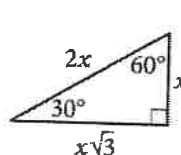
$$A = \ell w$$



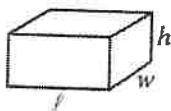
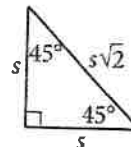
$$A = \frac{1}{2}bh$$



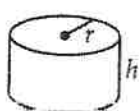
$$c^2 = a^2 + b^2$$



Special Right Triangles



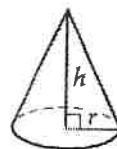
$$V = \ell wh$$



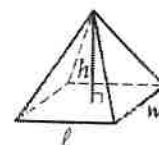
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

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- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

Thomas Realty Inc. purchased a property at \$250,000 in a certain town. The company received a 30% discount off the original price along with an additional 10% off the discounted price for purchasing the property within a week. Which of the following best approximates the original price, in dollars, of the property?

- A) \$8,333,000
- B) \$3,571,000
- C) \$750,000
- D) \$396,800

2

The perimeter of one face of a cube is $\frac{a}{4}$ inches, where a is a constant. Which of the following gives the volume of the cube, in cubic inches?

- A) $\frac{a^3}{4096}$
- B) $\frac{a^3}{512}$
- C) $\frac{a^3}{64}$
- D) a^3

3

An airplane is flying at an altitude of 15,000 feet from the ground. The pilot wants to land smoothly to the runway at an angle of depression of θ degrees. Which of the following equations can be used to find the horizontal distance, x , from the runway that the pilot should begin to descend?

- E) $x = \frac{15,000}{\sin \theta}$
- F) $x = \frac{15,000}{\tan \theta}$
- G) $x = \frac{15,000}{\cos \theta}$
- H) $x = 15,000 \cdot \tan \theta$

4

Which of the following expressions is equivalent to $\frac{x^2+10x+24}{x^3-x^2-20x}$?

- A) $\frac{x+6}{x-5}$
- B) $\frac{x+6}{x(x+4)}$
- C) $\frac{x+6}{x+4}$
- D) $\frac{x+6}{x(x-5)}$

5

The mean score of 9 players in a bowling game was 150 points. If the lowest individual score is dropped, the mean score of the remaining players becomes 160 points. What was the lowest score?

- A) 60
- B) 70
- C) 80
- D) 90

7

In isosceles triangle ABC, the measure of one angle is 80 degrees. Which of the following could be other angles in the triangle?

- I. 50 degrees.
- II. 20 degrees.
- III. 100 degrees.

- A) I only
- B) I and II only
- C) I and III only
- D) I, II, and III

6

$$N = 2 + t \cdot \frac{2}{3}$$

The total number of dark rings in the cross-section of wood represents the age of the tree in years. The equation above represents the number of rings, N , in the cross-section of a certain tree t years after it was planted. Which of the following best interprets the number $\frac{2}{3}$ in this context?

- A) The tree will gain 2 rings every 3 years of its age.
- B) The tree will grow 2 feet every 3 years of its age.
- C) The tree will lose $\frac{1}{3}$ ring every year of its age.
- D) The tree will gain 2 rings every $\frac{2}{3}$ years of its age.

8

An educational institution has three small groups by the ages of students. Twelve students less than 10 years old are placed in the class Yellow. Ten students at least 10 years old but less than 15 years old are placed in the class Blue. Thirteen students at least 15 years old are placed in the class Red. Which of the following could be the median of the ages, in years, of 35 students?

- A) 9
- B) 10
- C) 15
- D) 19

9

A statistics researcher wants to study the tendency of people to watch basketball games on television or go to the stadium to see the real game. The researcher visited the local pub on Sunday evening to ask 70 people who were watching sports in the pub. Which of the following statements best explains why this survey is NOT trustworthy?

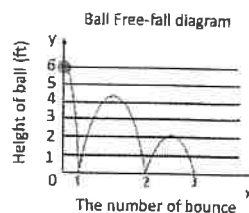
- A) The researcher should ask more people in the larger pub.
- B) This survey is biased in the place because the interviewees are likely to answer that they want to watch it on television in the pub.
- C) This survey is not trustworthy because the interviewees could be drunk when they were interviewed.
- D) This survey is biased because people in the pub usually couldn't afford to go to the stadium to see the real game.

10

Two points on a number line are both 4 units from the number -2. Which of the following absolute equations gives the coordinates of both points on the number line?

- A) $|x - 2| = 4$
- B) $|x + 2| = 4$
- C) $|x - 4| = 2$
- D) $|x + 4| = 2$

11



In the diagram above, a ball was dropped from the height of 6ft and bounced off the ground until it came to rest. After it was dropped from the height of 6ft, how many time was the ball at a height of 3ft?

- A) One
- B) Two
- C) Three
- D) Four

12

x	$f(x)$
4	0
2	-1
10	3

In the table above, some values of x and their corresponding $f(x)$ values are shown. If the function f is a linear, what is the value of $f(20)$?

- A) 10
- B) 8
- C) 6
- D) 4

13

A homeowners association invited community members in a meeting to discuss for the construction of children's library in the community. Of those invited to the meeting, 20% are HOA administrators, 60% were residents, the remaining 5 individuals were from the construction company. How many more residents were invited to the meeting than HOA administrators?

15

A group of friends decided to go to travel together and divided the \$1,200 cost of the trip package equally among themselves. However, two of the friends decided not to go to the travel later. If the remaining people still divide the cost equally and each share of the cost was increased by \$30, how many friends ended up going to travel?

14

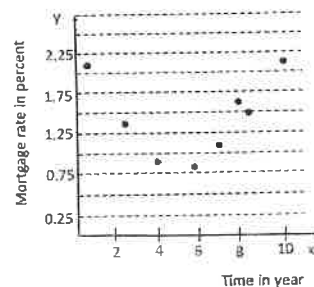
Masses of rare coins (grams)

Marcus	Francis
0.90	0.75
0.98	0.90
0.99	0.80
0.90	0.78
0.91	0.82
1.02	x

Marcus and Francis each collected rare coins while they were traveling other countries and the masses of the coins are shown in the table above. The mean of the coins collected by Marcus is 0.1 gram greater than the mean of the coins collected by Francis. What is the value of x ?

16

Mortgage rate in percent from 1991



The scatter plot above shows the mortgage rate, in percent, in a certain country from 1991 for a decade. Which of the following equations best represents the scatter plot?

- A) $y = -6(x - 5)^2 + 0.75$
- B) $y = 6(x - 5)^2 + 2.25$
- C) $y = 6(x - 5)^2 + 0.75$
- D) $y = 6(x + 5)^2 + 0.75$

17

$$f(x) = x^2 - 2x - 3$$

$$h(x) = a$$

In the system of equations above, a is a constant. If $h(x) \leq f(x)$ for all values of x , what is the maximum value of a ?

- E) 0
- F) 2
- G) -2
- H) -4

18

If the ratio of the number of teeth of two connected gears is $a:b$, then the ratio of the rotational speeds, in revolution per minute (rpm), of two gears is $b:a$. If gear A has 40 teeth and is rotated by the motor at a speed of 50 rpm and gear B has 80 teeth, what would the gear B's rotational speed, in revolution per minutes?

- A) 50
- B) 40
- C) 25
- D) 100

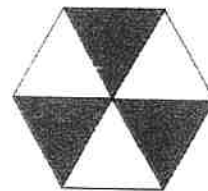
19

Matthew's driving records for a trip

	Distance (Miles)	Average Speed (mph)
Highway	120	60
Local	30	30

Matthew drove his car from his home to his friend's house in the other state. His driving records are shown in the table above assuming no traffic delay for the drive. If there is a traffic delay on the local and the travel time on the local roads will be expected to increase 40% due to slower traffic, what should be the average speed, in mph, on the highway in order to have the same total traveling time with no traffic delay?

20



In the pattern above, the regular hexagon consists of six equilateral triangles. If the area of one equilateral triangle is $4\sqrt{3} \text{ in}^2$, what is the perimeter, in inches, of the shaded regions?

21

$$a(2 - x) = 4x - 8$$

In the equation above, a is a constant. If the solution to the equation exists, which of the following statements must be true?

- I. If $a \neq -4$, then $x = 2$.
 - II. If $a = -4$, then the equation has infinitely many solutions.
 - III. If $a = -4$, then the equation has no solutions.
- A) I only
B) I and II only
C) I and III only
D) I, II, and III

22

The Survey results of two groups of samples for a new prototype

Sample	Percent in favor	Margin of error
A	72%	5.6%
B	57%	1.5%

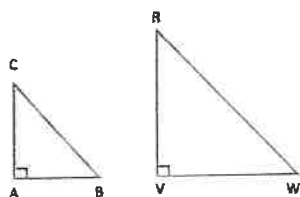
The distribution of the survey results of two groups of samples for a new prototype of the launching product is shown in the table above. The samples were selected at random from the same population. The margin of error for each sample were calculated using the same method. Which of the following could be a valid statement for the reason that sample A has a greater margin of error than that of sample B?

- A) Sample A had a larger sample size.
B) Sample A had a smaller sample size.
C) Sample A had a higher percent in favor.
D) Sample A had more unresponsive participants of the survey.

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

1



Triangle ABC is similar to triangle VWR as shown above. If $\cos \angle B = \frac{5}{13}$, what is the value of $\sin \angle R$?

- A) $\frac{12}{13}$
- B) $\frac{5}{13}$
- C) $\frac{12}{5}$
- D) $\frac{8}{13}$

2

The product of two positive integers is 204. If the first number is 5 less than the second number, what is the greater number?

3

At the special sale of headphones, the store offers 35% discount of the regular price. If the sales tax is 9% of the regular price and the customer paid \$12.50 for the sales tax, Approximately, what is the final sale price including the sales tax of the headphones?

- A) \$138.89
- B) \$102.78
- C) \$104.17
- D) \$90.28

4

$$\sqrt{(x-1)^2} = \sqrt{3x-3}$$

What is the largest solution to the given equation?

- A) -1
- B) 1
- C) 2
- D) 4

5

A sample of granite has a density of 2.7 g/cm^3 . If this sample shape is a cube, where the length of an edge is 1m , which of the following expressions correctly set up to find the mass, in kg, of this sample of granite?

- A) $\frac{2.7\text{g}}{\text{cm}^3} \times 100\text{cm}^3 \times \frac{1\text{kg}}{1,000\text{g}}$
B) $\frac{2.7\text{g}}{\text{cm}^3} \times 100^3\text{cm}^3 \times \frac{1,000\text{kg}}{1\text{g}}$
C) $\frac{2.7\text{g}}{\text{cm}^3} \times 100^3\text{cm}^3 \times \frac{1\text{g}}{1,000\text{g}}$
D) $\frac{2.7\text{g}}{\text{cm}^3} \times 1,000^3\text{cm}^3 \times \frac{1\text{kg}}{1,000\text{g}}$

6

Which of the following is an equation of a circle in the XY-plane if the circle has a center at $(1, -3)$ and passes through $(0, 0)$?

- A) $(x - 1)^2 + (y + 3)^2 = \sqrt{10}$
B) $(x - 1)^2 + (y + 3)^2 = 10$
C) $(x - 1)^2 + (y + 3)^2 = 20$
D) $(x + 1)^2 + (y - 3)^2 = 10$

7

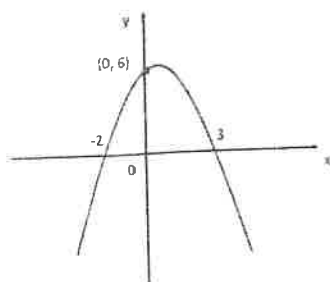
A circle has center O, and two points A and B are located on the circle. If $m\angle AOB = 68^\circ$, what is the measure of $\angle OAB$, in degrees?

- A) 112°
B) 56°
C) 68°
D) 34°

8

Math class A has 65 students and the mean score of class A is 83 percent. Math class B has 38 students and the mean score of class B is 95 percent. What is the mean score, in percent, of the combined classes A and B to the nearest tenth?

9



The graph of a quadratic function is shown in the XY -plane above. Which of the following is an equivalent form that could show y -intercept as a constant in the equation?

- A) $y = (x + 2)(-x + 3)$
- B) $y = -x^2 + x + 6$
- C) $y = -\left(x - \frac{1}{2}\right)^2 + \frac{25}{4}$
- D) $y = x(-x + 3) + 2(-x + 3)$

10

$$h(t) = -4.9t^2 + 14.7t$$

The equation h approximates the height of an object, in meters, t seconds after it was launched vertically straight up from the ground level. About how many seconds will the object take to hit the ground after it reached the maximum height?

- A) 1
- B) 1.5
- C) 2.5
- D) 3

11

$$P(3) = -7$$

In the polynomial function $P(x)$, the value of the function for one value of x is shown above. Which of the following must be true about $P(x)$?

- A) $x - 3$ is a factor of $P(x)$.
- B) $x + 3$ is a factor of $P(x)$.
- C) The remainder is -7 when $P(x)$ is divided by $x - 3$.
- D) $(-3, -7)$ is one of the solutions of the polynomial function $P(x)$.

12

Region	# of clam shells	Region	# of clam shells
A	22	F	3
B	18	G	27
C	19	H	18
D	25	I	22
E	15	J	26

Ocean scientists studied about the radiation effect on the ocean life from the near nuclear power plant. A square area for their research in the mudflats near the nuclear power plant measures 10 feet by 10 feet. The ocean scientists randomly chose 10 regions, A through J in the research location as shown; each region measures 1 foot by 1 foot and they counted the number of clam shells, found in each region. Which of the following is a reasonable approximate number of clam shells in the entire square area (10 ft by 10 ft)?

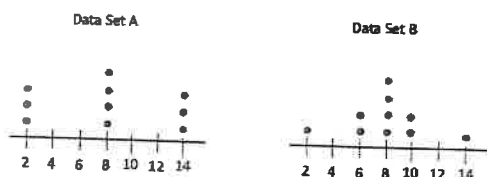
- A) 19.5
- B) 195
- C) 1,950
- D) 19,500

13

If the height of an isosceles right triangle, when the base is the hypotenuse of the isosceles right triangle, is $2\sqrt{2}$ inch, what is the area of the isosceles right triangle, in square inches?

- E) 2
- F) $2\sqrt{2}$
- G) 4
- H) $4\sqrt{2}$

14



The dot plots for data set A and B are shown above. Which of the following statements must be valid?

- I. The median value of data set A and B are equal.
- II. The mean value of data set B is higher than the mean value of data set A.
- III. The standard deviation of data set A and B are not the same value.

- A) I only
- B) I and II
- C) I and III
- D) I, II, and III

15

$$x^2 - 10x + k = 0$$

The solutions to the equation above, where k is a constant, are $x = 5 \pm \sqrt{7}$. What is the value of k ?

16

$$\begin{aligned} -py - \frac{5}{3}x &= \frac{1}{3}(x + 1) \\ \frac{2}{3}x - \frac{1}{2} &= -\frac{1}{2}y + 3 \end{aligned}$$

In the system of equations, p is a constant. If the system has no solutions, what is the value of p ?

17

The measure of angle P is $\frac{2}{5}\pi$ radians less than the measure of angle R. How much greater is the measure of angle R than the measure of angle P, in degrees?

19

What is the y-intercept of the exponential graph of $y = k(2)^{\frac{x}{3}} - l$ in the xy-plane, where k and l are constants?

- A) $(0, k - l)$
- B) $(k - l, 0)$
- C) $(0, -l)$
- D) $(-l, 0)$

18

Swimming instructors and students will have swim lessons in the swimming pool. Each instructor will lead a group of no more than 5 students. The maximum capacity of the swimming pool is 120. Which of the following systems best describes the possible numbers of instructors and students in the swimming pool if x represents the number of swimming instructors and y represents the number of students?

- A) $\begin{cases} x + y < 120 \\ 5x > y \end{cases}$
- B) $\begin{cases} x + y < 120 \\ 5x < y \end{cases}$
- C) $\begin{cases} x + y \leq 120 \\ 5x \geq y \end{cases}$
- D) $\begin{cases} x + y \leq 120 \\ 5x \leq y \end{cases}$

20

Program at the gym	Year		
	2000	2001	2002
Kickboxing	53	64	85
Aerobic	35	40	44
Zumba	67	88	103
Cycling	15	20	26

Which of the following best represents the average rate of change in the annual number of registrations for Zumba program at the gym from the year 2000 to 2002?

- A) 15 per year
- B) 16 per year
- C) 17 per year
- D) 18 per year

21

$f(x)$ equals 330% of x

For $x > 0$, the function f is defined as the above.
Which of the following could describe the function f ?

- A) Linearly increasing
- B) Linearly decreasing
- C) Exponentially increasing
- D) Exponentially decreasing

22

The graph of $2x - 7y - 3 = 0$ is translated 3 units up in the XY -plane. What is the coordinate of x -intercept?

- A) $(-9, 0)$
- B) $(9, 0)$
- C) $(3, 0)$
- D) $(-3, 0)$

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

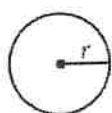
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

Unless otherwise indicated:

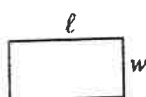
- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

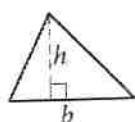


$$A = \pi r^2$$

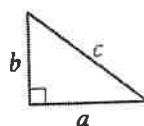
$$C = 2\pi r$$



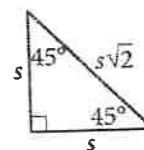
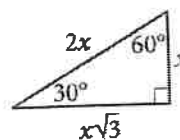
$$A = \ell w$$



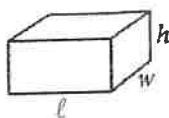
$$A = \frac{1}{2}bh$$



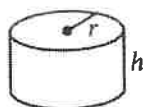
$$c^2 = a^2 + b^2$$



Special Right Triangles



$$V = \ell wh$$



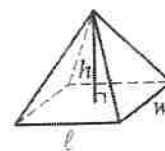
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction ($\frac{7}{2}$) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1

Adrian receives monthly allowance from his parents. He started to deposit some fixed amount of money from his allowance into his piggy bank every month. The function $f(t) = 125 + 35t$ gives the total amount of money he saved, in dollars, in his piggy bank after t monthly deposits. What could 125 represent in this context?

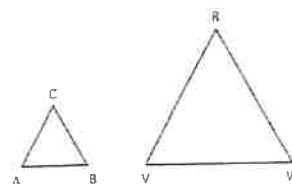
- A) Monthly deposit from his allowance.
- B) Total amount of money he saved after t months.
- C) The amount of money he saved already before the first deposit.
- D) The price of his piggy bank when he purchased.

2

A customer purchased 5 shower curtains that were each the same price. She used a coupon for 20% off the entire purchase. She paid \$145 in total after using the discount coupon. If the sales tax was ignored, what was the original price of one shower curtain?

- A) \$36.25
- B) \$23.20
- C) \$145.00
- D) \$29.00

3



In two triangles ABC and VWR above, if $\angle A \cong \angle V$, what other pieces of information are sufficient to prove whether two triangles are similar?

- I. $\angle C \cong \angle R$
- II. $\frac{AC}{VR} = \frac{AB}{VW}$

- A) I only
- B) II only
- C) Either I or II
- D) None

4

$$30m + 45w = 480$$

The equation given above describes the relationship between the number of men, m , and the number of women, w , that can be serviced for haircut at a local haircut shop on a given day. If the business gave services for 6 women on a certain day, how many men could the business give the haircut service on this day?

5

For a certain triangular region, the ratio of the height to its base is always a constant. If the ratio of the height to its base is 3:5 and the height of the triangular region is increased by 9, by how much does the base of the region need to be increased to maintain the ratio?

- A) 9
- B) 12
- C) 13
- D) 15

6

Circle A has a radius length of k inches. Circle B has a circumference that is 10π inches greater than the circumference of circle A. The function h gives the area of circle B, in square inches. Which of the following defines h correctly?

- A) πk^2
- B) $\pi(k + 5)^2$
- C) $\pi(k + 10)^2$
- D) $\pi(k + 2)^2$

7

$$g(x) = 1,200(1.22)^x$$

The function $g(x)$ models the value, in dollars, of a certain stock at the end of the year, where x is the number of years after the stock was bought in 2000. Which of the following best interprets the meaning of " $g(3)=2,179.02$ " in this context?

- A) The value of the stock will be approximately \$2,179.02 by the end of 2003.
- B) The value of the stock will be increased by \$2,179.02 by the end of 2003.
- C) The value of the stock will be increased by \$3 by the end of 2003.
- D) The value of the stock will be increased by \$2,179.02 every three years from 2000.

8

$$2\sqrt{3+p} = \frac{4x+w}{3w}$$

The equation above relates three distinct positive real numbers p , x , and w . Which of the following correctly solve for w in terms of x and p ?

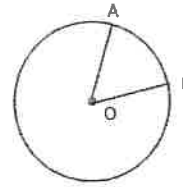
- A) $\frac{4x}{3(1+\sqrt{3+p})}$
- B) $\frac{4x}{6\sqrt{3+p}-1}$
- C) $\frac{4x}{1-6\sqrt{3+p}}$
- D) $\frac{4x}{1+6\sqrt{3+p}}$

9

A triangle has length of three sides; $2\sqrt{3}$, $6\sqrt{2}$, and $2\sqrt{21}$ inches. What is the area of the triangle, in square inches?

- A) $2\sqrt{6}$
- B) $3\sqrt{6}$
- C) $4\sqrt{6}$
- D) $6\sqrt{6}$

11



In the circle O above, the measure of central angle AOB is 30° . What is the measure, in radians, of its associated arc AB?

- A) $\frac{\pi}{3}$
- B) $\frac{\pi}{4}$
- C) $\frac{\pi}{6}$
- D) $\frac{\pi}{2}$

10

The quadratic expression $5x^2 + ax - 68$, where a is a constant, can be factored as $(mx + k)(x + l)$, where m , k , and l are integers. Which of the following must be an integer?

- A) $\frac{6}{m}$
- B) $\frac{68}{m}$
- C) $\frac{m}{k}$
- D) $\frac{68}{l}$

12

$$8\sqrt[3]{27x^{81}} \cdot \sqrt[7]{2^{14}x^{21}}$$

The expression above is equivalent to kx^m , where k and m are constants and $x > 1$. What is the value of $\frac{k}{m}$ in the simplest fraction?

- A) $\frac{4}{5}$
- B) $\frac{16}{8}$
- C) $\frac{8}{5}$
- D) $\frac{16}{5}$

13

$$\begin{aligned}y &= 3x^2 - x - 2 \\ y &= 2x - k\end{aligned}$$

In the system of equations above, k is a constant. The graphs of the system will intersect at exactly one point, (x, y) , in the XY -plane. What is the value of k ?

- A) $\frac{11}{4}$
- B) $-\frac{11}{4}$
- C) $\frac{25}{12}$
- D) $-\frac{25}{12}$

14

An isosceles right triangle has $6\sqrt{2}$ cm as the length of the hypotenuse. What is the area, in square cm, of the isosceles right triangle?

15

$$f(x) = \frac{2(x-1)}{(x-1)(x+2)}$$

In the rational expression above, for what value of x is the function f undefined?

- I. $x = 1$
- II. $x = -2$
- III. $x = -1$

- A) I only
- B) II only
- C) I and II
- D) II and III

16

A parabola has a vertex at $(-2, 5)$ and intersects the x -axis twice in the XY -plane. If the equation of parabola can be written as $y = ax^2 + bx + c$, where a , b , and c are constants. Which of the following could be $a + b + c$?

- A) 5
- B) 6
- C) 7
- D) 0

17

A sample was selected at random to estimate the proportion of a population for a certain characteristic. The estimated proportion of the population for a certain characteristic is 0.78, with an associated margin of error of 0.02. Based on the information given, what can you conclude about the proportion of the population regarding the characteristic surveyed?

- A) The proportion of the population for the characteristic is exactly 0.78.
- B) It is plausible that the proportion of the population for the characteristic is between 0.76 and 0.80.
- C) It is plausible that the proportion of the population for the characteristic is greater than 0.80.
- D) It is plausible that the proportion of the population for the characteristic is less than 0.76.

18

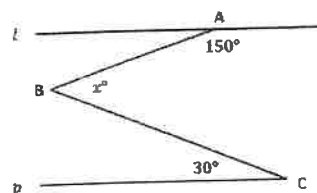
A truck driver wanted to figure out how many cargo containers can load on his truck to drive safely in a bridge. The bridge allows a truck that weighs no more than 5,000 pounds, including all loads on the truck. What is the maximum number of cargo containers allowed to load on the truck if the weight of truck with no loads is 1,200 pounds and each cargo container weighs 150 pounds?

- A) 24
- B) 25
- C) 26
- D) 27

19

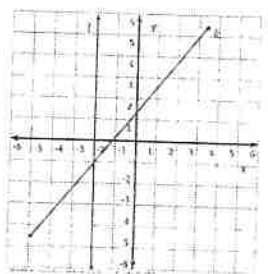
If $2x + k$ is a factor of $2x^3 + 13x^2 - 7x$, where k is an integer, what is the value of $|k|$?

20



In the figure above, $l \parallel p$. What is the measure of angle x ?

21

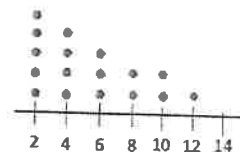


The system of equations is graphed above. If a new graph of $4x - y + 7 = 0$ is also graphed in the same XY-plane, how many solutions (x, y) will the system of three equations have?

- A) One
- B) Two
- C) Three
- D) Infinitely many

22

Data Set A



There are 17 values in the dot plot in the Data set A above. If one more number 14 is added to the Data set A, which of the following is valid statement based on the dot plots.

- I. The median is unchanged.
 - II. The mode is unchanged.
 - III. The range is unchanged.
- A) I only
 - B) I and II only
 - C) I and III only
 - D) II only

STOP

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Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

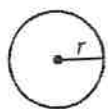
The questions in this section address a number of important math skills.
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NOTES

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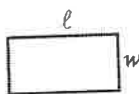
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- Figures provided are drawn to scale.
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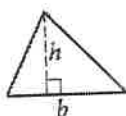


$$A = \pi r^2$$

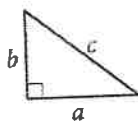
$$C = 2\pi r$$



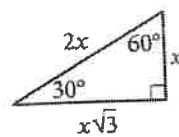
$$A = \ell w$$



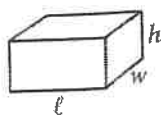
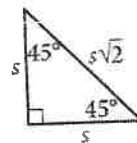
$$A = \frac{1}{2}bh$$



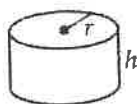
$$c^2 = a^2 + b^2$$



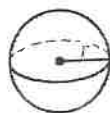
Special Right Triangles



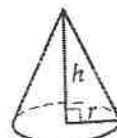
$$V = \ell wh$$



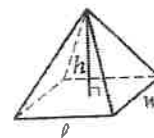
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

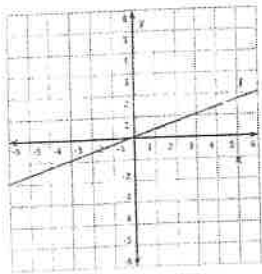
The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

1



The graph of a line l is shown above in the XY -plane. The slope of a line k is twice the slope of the line l . If the graph of a line k passes through a point $(3, 6)$, what is the coordinates on a line k in the XY -plane?

- A) $(-3, 3)$
- B) $(0, 0)$
- C) $(-3, 4)$
- D) $(-3, 2)$

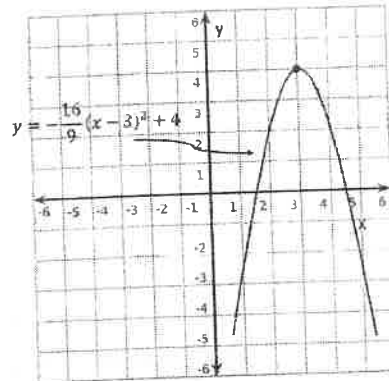
2

$$y = 3x^2 - k$$

In the equation above, k is a positive constant. Which of the following is an equivalent form of the given equation?

- A) $y = (3x - \sqrt{k})(x + \sqrt{k})$
- B) $y = (\sqrt{3x} - \sqrt{k})(\sqrt{3x} + \sqrt{k})$
- C) $y = (\sqrt{3x} + \sqrt{k})(\sqrt{3x} - \sqrt{k})$
- D) $y = (\sqrt{3x} - k)(\sqrt{3x} + k)$

3



The quadratic graph above models the daily profit (y value), in hundred dollars, that a vendor expects to make from selling churros for a unit price of x dollar. Based on the model, what is the unit profit, in dollars, of churros to make the same profit when the vendor sells it for a unit price of 4 dollars?

4

Isaiah works as sales associate at a retail store. He is paid 15% of the sales he made. He also gets a bonus which is additional 10% of his pay if his sales is over \$1,000 during a pay period. If he earns a total of \$ 587.40 including a bonus on his pay period, what was the amount of the sales, in dollars, he made during one pay period?

5

$$C(x) = 350 + kx$$

A manufacturing company's total cost, in dollars, to produce x calculators is given by the function $C(x)$ above, where k is a constant. The total cost, in dollars, to produce 200 calculators is \$5,350. What is the total cost, in dollars, to produce 250 calculators?

6

The total population, P , of rabbits in a certain area can be modeled by a quadratic function that is defined in terms of t , where t is the time in months. At a time of 5 months, the total population of rabbits is 300, and at a time of 12 months, the total population of rabbits is 1,252. If the initial population of rabbits is 100, then what is the total population of rabbits in this area after 2 years?

- A) 132
- B) 13,200
- C) 4,708
- D) 4,608

7

Score	Score frequencies	
	Class A	Class B
5	2	0
10	0	0
15	4	8
20	2	4
25	7	2
30	0	2
Total	15	16

For a certain math game, players can score from 0 to 30 for 6 games. If a player wins the game, the person gains 5 points but if a player loses the game, no point gains or loses. The table shows the distribution of the scores of 31 players in two classes. How much is the median of class B is greater than the median of class A?

- A) 0
- B) 2.5
- C) 5
- D) 7.5

8

$$x^2 - 8x + k = 0$$

In the quadratic equation above, k is a constant. The equation has two real solutions if $k < m$. What is the least possible value of m ?

9

Max is folding paper cranes for a friend's birthday party. He plans to make 2,000 paper cranes to decorate the reception table. If it takes 5 minutes to fold one paper crane and he already made k paper cranes, which of the following expressions represents the number of hours needed for him to complete the rest of work?

- A) $5(2,000 - k)$
- B) $12(2,000 - k)$
- C) $\frac{1}{12}(2,000 - k)$
- D) $\frac{1}{12}(k - 2,000)$

10

A circle is graphed in the xy -plane and the endpoints of the diameter of a circle are $(2, 0)$ and $(2, -4)$. How many points will the circle intersect with y -axis?

- A) 0
- B) 1
- C) 2
- D) 3

11

Which of the following expressions is equivalent to $(-3a^{\frac{1}{2}})^{\frac{2}{3}}$, where $a > 0$?

- A) $6\sqrt[3]{3a}$
- B) $\sqrt[3]{9a}$
- C) $-\sqrt[3]{9a^2}$
- D) $-\sqrt[3]{9a}$

12

$$\begin{aligned} 2x + 4 &= k \\ 5x - 3y &= 4 \end{aligned}$$

In the system of equations above, k is a constant. What is the value of y of the solution to the system in terms of k ?

- A) $\frac{28-5k}{6}$
- B) $\frac{5k-28}{6}$
- C) $\frac{5k+4}{6}$
- D) $\frac{5k-4}{6}$

13

$$\begin{aligned}20 \leq a \leq 80 \\ 60 \leq b \leq 130\end{aligned}$$

The intervals of a and b are shown above. Which of the following inequalities represents correctly for the interval of c if $c = b - a$?

- A) $|45 - c| \leq 65$
- B) $|c - 45| \geq 65$
- C) $|c - 45| \leq 5$
- D) $|c - 45| \geq 5$

14

$$f(t) = \sqrt[3]{2 - 2x^2}$$

In the function above, what values of x are defined for which $f(x)$ is a real number?

- A) $-1 \leq x \leq 1$
- B) $-\sqrt{2} \leq x \leq \sqrt{2}$
- C) $-2 \leq x \leq 2$
- D) *all real numbers*

15

$$\begin{aligned}f(x) &= (x - 3)(x + 7) \\ h(x) &= x^2\end{aligned}$$

In the quadratic functions above, what change could be made from $h(x)$ to be $f(x)$?

- A) $h(x)$ was translated 2 units right and 25 units down.
- B) $h(x)$ was translated 2 units left and 25 units down.
- C) $h(x)$ was translated 2 units left and 25 units up.
- D) $h(x)$ was translated 2 units right and 25 units up.

16

Thomas wants to share 1.5 gallons of orange juice with his friends. He distributed 8 fluid ounce cups to 7 friends and himself. How many of full 8 fluid ounce cups of orange juice each person would get if 1.5 gallons of orange juice was divided equally to all? (1 gallon = 128 fluid ounces)

- A) 2
- B) 3
- C) 4
- D) 5

17

During the end-of-year sale, Elon bought a electric car at a 20% discount. He paid a total of k dollars, which included the discounted price and a 9.5% sales tax on the discounted price. What is the original price of the electric car before the discount in term of k ?

- A) $\frac{k}{(0.8)(1.095)}$
 B) $\frac{k}{(0.2)(0.095)}$
 C) $\frac{k}{(0.8)(0.095)}$
 D) $k(0.8)(1.095)$

18

$$f(x) = x^2 - 4x - 5$$

The quadratic function is shown above. Which of the following equivalent forms of the function shows the minimum value of the function as constant?

- A) $f(x) = x(x - 4) - 5$
 B) $f(x) = (x - 2)^2 - 9$
 C) $f(x) = (x - 5)(x + 1)$
 D) $f(x) + 5 = x(x - 4)$

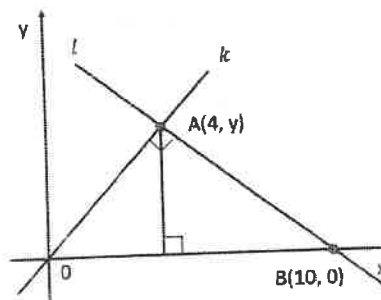
19

$$h(x) = 2x^4 + 4$$

In the xy -plane, the equation $y = h(x)$ is graphed. Which of the following could be the value of $h(x)$?

- A) 0
 B) 1
 C) 2
 D) 4

20



Lines k and l are shown in the xy -plane above. If two lines are perpendicular and intersect at point A , what is the area of triangle OAB ?

- A) 20
 B) $5\sqrt{6}$
 C) $10\sqrt{3}$
 D) $10\sqrt{6}$

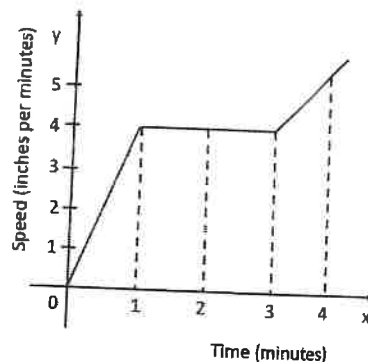
21

$$f(x) = 3x^2 - 12$$
$$f(x - k) = 3x^2 - 12x$$

If the first function $f(x)$ is translated k units horizontally, the function will be $3x^2 - 12x$ as shown above. What is the value of k ?

- A) 2
- B) -2
- C) 0
- D) -1

22



The graph above models the speed, y , in inches per minutes, of a snail during the first 4 minutes of travel time, x . What is the total distance traveled by the snail between 1 min to 3 min?

- A) 16 inches
- B) 12 inches
- C) 10 inches
- D) 8 inches

STOP

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Math

22 QUESTIONS
(TIME: 35 MIN)

DIRECTIONS

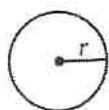
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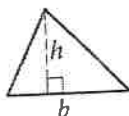


$$A = \pi r^2$$

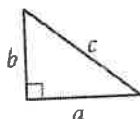
$$C = 2\pi r$$



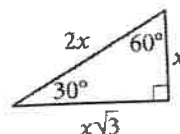
$$A = \ell w$$



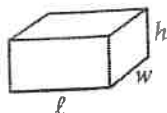
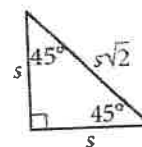
$$A = \frac{1}{2}bh$$



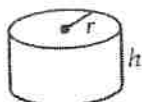
$$c^2 = a^2 + b^2$$



Special Right Triangles



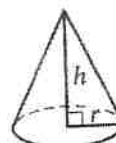
$$V = \ell wh$$



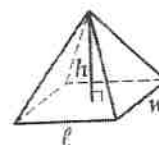
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

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The sum of the measures in degrees of the angles of a triangle is 180.

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1

A rowing team entered a 500-meter race. The team rowed at an average speed of 32 strokes per minutes. If the team took 350 seconds to complete the 500-meter race, how many strokes did it take the team during the race to the nearest whole number?

2

The measure of an angle is $\frac{2}{3}\pi$ radians. If it is converted into degrees, it has a measure of m degrees. What is the value of m ?

3

The value of function f will increase by 30% for every value of x increase by 1 in the function. Which of the following could be the function f ?

- A) $f(x) = 2(0.3)^x$
- B) $f(x) = 23(1.3)^{x+2}$
- C) $f(x) = 0.13(1.3)^{-x}$
- D) $f(x) = 25(1.3)^{2x}$

4

$$f(x) = (x - 5)(x + 2)^2$$

The function f is defined above. If $f(x - 2) = 0$, what could be the value of x ?

- A) 0
- B) 1
- C) 2
- D) 5

5

Participants in the international model airplane competition need to build a light-weight (less than 30 pounds) airplane model and a heavy-weight (more than or equal to 30 pounds) airplane model for the competition. A heavy-weight airplane model uses 2 small wheels and 4 large wheels, and a light-weight airplane model uses 1 small wheel and 2 large wheels for their landing gears, respectively. There are 40 large wheels and 30 small wheels available in the competition, and each participant must build one heavy-weight airplane model and one light-weight airplane model. What is the maximum number of participants that could attend the competition?

6

Nutritional information for 1 ounce servings of seeds

	Total fat (grams)	Protein (grams)	Calories
Barley	0.65	3.54	100
Brown rice	0.26	0.73	31

The table above shows nutritional information for 1-ounce of barley and brown rice. How many more grams of total fat are in one pound of barley than are in one pound of brown rice?
(1 pound = 16 ounces)

- A) 2.81
- B) 33.72
- C) 0.39
- D) 6.24

7

Jimmy bought a \$90 daily pass for unlimited rides in magic mountain theme park. The average price for a ride in the theme park is \$6.50. If he took 8 different rides in the morning, at least how many rides does he need to take in the afternoon in order to save money in the cost?

- A) 4
- B) 5
- C) 6
- D) 7

8

$$x^6 - y^6 = 16$$

$$x^3 - y^3 = 8$$

In the two equations above, what is the value of $x^3 + y^3$?

9

Sarah surveyed 80 people who visited in a local library and asked if they are in favor of the proposal to use \$150,000 of local taxes for upgrading desks and chairs for the libraries in the city. She found that 60 of those surveyed were in favor of the proposal. Which of the following statements is the most reasonable conclusion?

- A) This sampling method was flawed and could be biased because of the survey place.
- B) No prediction should be made because the sample size is too small.
- C) 75% of the votes will be in favor of the proposal when the actual votes were taken.
- D) The result of this survey is valid, unbiased, and must forecast the actual votes fairly.

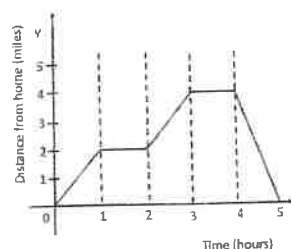
10

Distribution of tossing two coins

		Quarter	
		Heads	Tails
Nickel	Heads	12	9
	Tails	10	11

The two-way table shows the distribution of the results of tossing two coins. For what percent of the tosses were the Quarters landed on tails to the nearest tenth?

11



Charles walked from home to his friend's house. He stopped by a restaurant on the way to have a lunch for an hour and walked again to reach his friend's house and stayed there with his friend for an hour and finally he ran straight back to his home on the same road. The graph above show distance from his home by hours. How many times faster in speed, in miles per hour, was he running back home than walking to his friend's house?

- A) 1.5
- B) 2.0
- C) 2.5
- D) 4.0

12

The graphs of the following quadratic equations each have x-intercepts of -3 and 7 in the XY-plane. Which equation has its vertex farthest from the x-axis?

- A) $y = \frac{1}{2}(x + 3)(x - 7)$
- B) $y = -\frac{1}{5}(x + 3)(x - 7)$
- C) $y = 6(x + 3)(x - 7)$
- D) $y = -12(x + 3)(x - 7)$

13

With greater depth of the atmosphere, more air is pressing down from above. Therefore, air pressure is greatest at sea level and falls with increasing altitude. Michael hiked the Mt. Everest in winter and the air pressure dropped at a constant rate of 0.768 kPa per every 100 meters vertical distance. The height of Mt. Everest is about 8.85 kilometers from the sea level. Which of the following is closest to the total drop in air pressure, in psi (pounds per square inch), over the course of the entire hike from the sea level to the top of the mountain? (1 kPa = 0.145 psi)

- A) 0.99
- B) 468.7
- C) 68.0
- D) 9.86

14

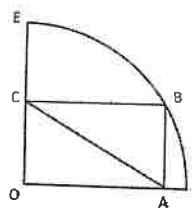
If $\frac{3x}{y} = 7$, what is the value of $\left(\frac{4xy}{3}\right) \cdot (2y)^{-2}$?

15

Which of the following describes exponential relationship between two variables listed?

- A) Airplane's speed v , in feet per minute, will decrease at a constant rate of 1,500 feet per minute every minute m .
- B) The depth h of water in a water container decreases by 10 inches each minute m as the water is going out at a constant rate.
- C) The air pressure p decreases by 7.68 Pa when the height h increases every meter.
- D) The sales of a product s , in dollars, in a company is expected to grow 20% every year t for the next decade.

16



Rectangle OABC is inscribed into the quarter circle O as shown above. The length of OA is $5\sqrt{3}$ and the measure of $\angle OAC$ is 30° . What is the area of the quarter circle O?

- A) 100π
- B) 50π
- C) 25π
- D) 20π

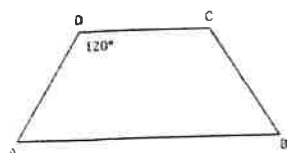
17

Science classes enrollment at Valencia High School

	Male	Female	Total
Physics	65		100
Chemistry			
Total	110		200

The table above shows the distribution of science classes (Physics and Chemistry) at Valencia high school partially. After completing the table, what percent of female students are taking Chemistry class to the nearest whole number?

18



An isosceles trapezoid ABCD is formed by cutting the regular hexagon into half as shown above and the perimeter of trapezoid ABCD is 10, what is the area of the trapezoid?

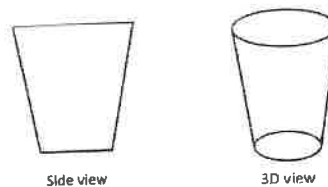
- A) $3\sqrt{3}$
- B) $5\sqrt{3}$
- C) $6\sqrt{3}$
- D) 6

19

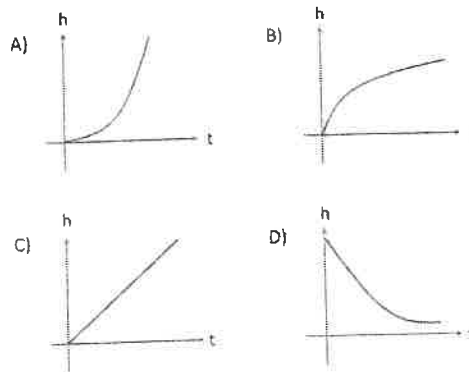
Which of the following transformation should be made for the graph of $f(x) = (x - 2)^2$ to become the graph of $g(x) = -(x - 2)^2 + 3$?

- A) Translate 3 units left.
- B) Reflect over x-axis and translate 3 units right.
- C) Reflect over y-axis and translate 3 units up.
- D) Reflect over x-axis and translate 3 units up.

20

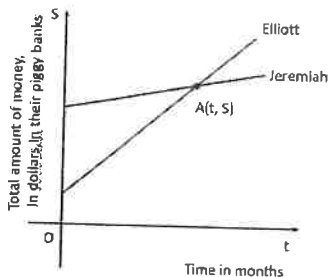


Two different views for a water cup are shown above. Assuming that water is being poured at a constant rate, which of the following best represents the height of water h in the water cup as time t elapses in the x-axis?



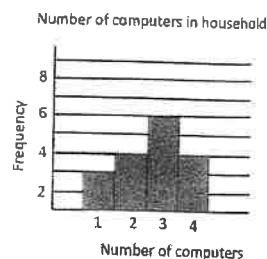
21

Jeremiah has \$120 in his piggy bank and deposits \$30 every month of his allowance from parents. Elliott has \$40 in his piggy bank and deposits \$50 every month from his part-time job. The linear graphs for this situation are shown below.

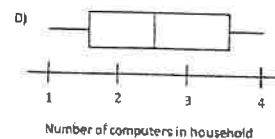
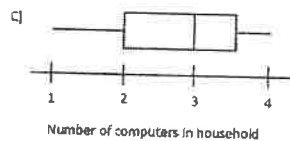
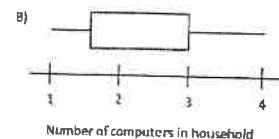
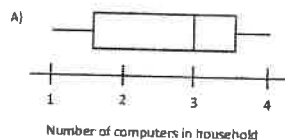


If two lines are graphed in the tS -plane. What is the value of S , in dollars, when they have the same amount of money in their piggy banks?

22



A teacher surveyed her class for the number of computers in the household. The results are shown in the bar graph above. Which of the following box plots best represents her survey result?

**STOP**

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Math

22 QUESTIONS

(TIME: 35 MIN)

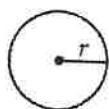
DIRECTIONS

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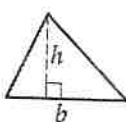
REFERENCE

$$A = \pi r^2$$

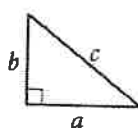
$$C = 2\pi r$$



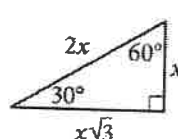
$$A = \ell w$$



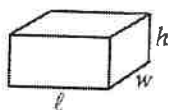
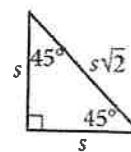
$$A = \frac{1}{2}bh$$



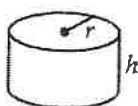
$$c^2 = a^2 + b^2$$



Special Right Triangles



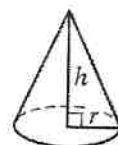
$$V = \ell wh$$



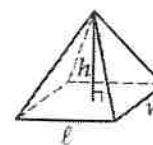
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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1

$$(x + 1)^2 + (y - 2)^2 = 64$$

In the circle equation above, if the circle is translated 3 units right in the XY -plane, which of the following represents the equation of the new circle?

- A) $(x - 4)^2 + (y - 2)^2 = 64$
- B) $(x - 2)^2 + (y - 2)^2 = 64$
- C) $(x + 4)^2 + (y - 2)^2 = 64$
- D) $(x + 1)^2 + (y + 1)^2 = 64$

2

Characteristics of customers at a Hotel in Las Vegas

	Buffet	No Buffet	Total
Gambling	650	150	800
No Gambling	190	20	200
Total	830	170	1,000

The table shows a distribution of characteristics on a certain day for 1,000 customers at a hotel in Las Vegas. The table shows the number of customers who did gambling, visited buffet restaurant, both, or neither. Based on the data, what is the probability that a customer, if selected at random, didn't visit buffet restaurant given that the customer did gambling?

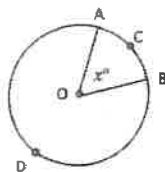
3

Two identical cylinders, each has a radius of 2 inches on the circular bases and h inches on their heights, are attached together along the circular bases. The volume of the new cylinder is $k\pi$ cubic inches. If the surface area of one cylinder before attached is also $k\pi$ square inches, what is the value of height (h), in inches, of one cylinder?

4

If 540 is $k\%$ less than 2,000, what is the value of k ?

5



In the circle O above, the length of a small arc $\widehat{ACB}$ is 3π and $x = 45$. What is the area of large sector ADB?

- A) 126π
- B) 21π
- C) 18π
- D) 6π

7

$$(x + a)(bx - 3) = 4x^2 - 5x - 6$$

The equation above is true for all x , where a and b are constants. What is the value of ab ?

- A) 4
- B) -4
- C) 8
- D) -8

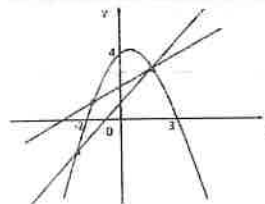
6

$$P = 92t + 80$$

A plumber charges for a job as shown above. The equation gives the total cost, in dollars, of a job that takes t hours. Two customers A and B hired this plumber for their jobs and the plumber worked 3 hours longer for customer B's job. How much more did the plumber charge, in dollars, on customer B than customer A?

- A) 80
- B) 92
- C) 184
- D) 276

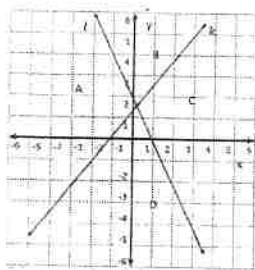
8



A system of three equations is graphed in the XY -plane above. How many solutions does the system have?

- A) Zero
- B) One
- C) Two
- D) Three

9



$$\begin{aligned} 5y - 6x &\leq 7 \\ 3x + 2y &\geq 3 \end{aligned}$$

The system of inequalities is graphed in the XY -plane. Which region in A through D is the solution located based to the system above?

- A) A
- B) B
- C) C
- D) D

10

$$\sqrt{\frac{x+4}{2}} = -3x$$

In the equation above, what is the set of all possible solutions to the equation?

- A) $\left\{-\frac{4}{9}, \frac{4}{9}\right\}$
- B) $\left\{-\frac{4}{9}\right\}$
- C) $\left\{\frac{4}{9}\right\}$
- D) No solution.

11

Shipping Cost in local UPS

Weight (pounds)	Shipping Cost (dollars)
10	18.30
20	30.60
40	55.20

The table above shows the shipping cost, in dollars, in local UPS of packages by its weight, in pounds. If there is a linear relationship between the shipping cost and the weight of the packages, which of the following can be used to determine the total shipping cost (y), in dollars, of packages with a weight of x pounds?

- A) $y = 1.23x$
- B) $y = 1.23x + 6$
- C) $y = 12.3x + 0.6$
- D) $y = 12.3x + 6$

12

In Peter's math class, the class average (arithmetic mean) for math unit test was 85 percent for 36 students. However, three of his class students dropped the course. So, the teacher recalculated the mean for the remaining class and figured that the new mean was 87 percent. What was the mean score for three student who dropped the course?

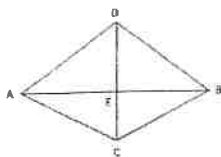
- A) 61
- B) 62
- C) 63
- D) 64

13

Nathan purchased an old rare coin in auction at \$180. He found out that the value of the coin increases 15% of its value the previous year. The estimated value of the coin, in dollars, 3 years after purchase can be written as $180k$, where k is a constant. What is the approximate value of k ?

- A) 1.32
- B) 1.52
- C) 1.15
- D) 0.0034

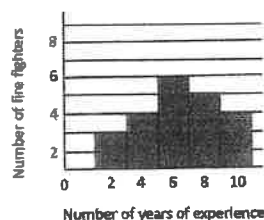
14



In the figure above, $AD = BD$, $AC = BC$. If $CD = 7$ and $AE = 4$, what is the area of kite ACBD?

- A) 28
- B) 30
- C) 56
- D) 60

15



The bar graph above shows the distribution of the number of years of experience for 22 fire fighters who have served in a certain local fire department. If a new fire fighter who has 6 years of experience joins the local fire department, what will be the effect on the mean and median of the data set?

- A) The mean and median will remain the same.
- B) The mean will decrease but the median will increase.
- C) The mean will increase but the median will decrease.
- D) The mean will decrease but the median will remain the same.

16

The expression $x^2 + kx + 20$ is equivalent to $(x - 2)^2 + m$, where k and m are constants. What is the value of $-\frac{m}{k}$?

17

$$y = ax(x^2 - 1)(x^2 - 4)$$

In the equation above, a is a constant. How many distinct x -intercepts does the graph of the equation in the XY -plane have?

- A) Five
- B) Four
- C) Three
- D) Two

18

Let $y = f(x)$ is a linear function with a negative slope and graphed in the XY -plane. Which of the following must be true?

- I. $f(a) > f(b)$ if $a > b$
- II. $f(b) < 0$ if $b > 0$
- III. $f(a) < 0$ if $a < 0$

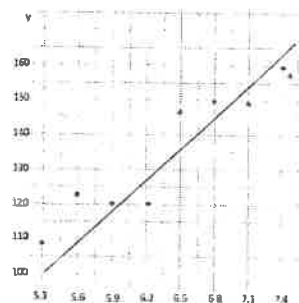
- A) I only
- B) I and II only
- C) I and III only
- D) None

19

Kevin walked three fourth of the distance from his home to the library. For the rest of the way to the library, he ran 2 times as fast as he walked. If he took 30 minutes to walk three fourth of the way, how much time, in minutes, did it take him from his home to the library?

- A) 32
- B) 35
- C) 40
- D) 45

20



The scatter plot above shows the distribution of heights (y values) of women based on their shoe sizes (x values). The line of best fit is also drawn in the XY -plane. What is the number of women for which the line of best fit predicts a value less than the actual value?

21

Ashley has a budget of \$50 to spend on notebooks and binders for her school. Notebooks cost \$2.50 each and binders cost \$5.25 each. Which of the following inequalities represents this situation, where n is the number of notebooks she can purchase, and b is the number of binders she can purchase? (Assume there is no other costs)

- A) $2.50b + 5.25n = 50$
- B) $2.50b + 5.25n \leq 50$
- C) $2.50n + 5.25b > 50$
- D) $2.50n + 5.25b \leq 50$

22

What is the y -intercept of the exponential graph of $y = 2k \left(\frac{1}{2}\right)^{\frac{x}{3}+1}$ in the XY -plane, where k is a constant?

- E) $(0, k)$
- F) $(k, 0)$
- G) $\left(0, \frac{k}{2}\right)$
- H) $\left(-\frac{k}{2}, 0\right)$

STOP

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Math

22 QUESTIONS

(TIME: 35 MIN)

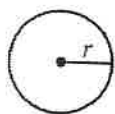
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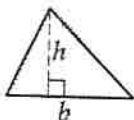
REFERENCE

$$A = \pi r^2$$

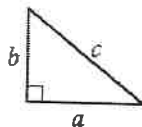
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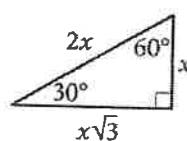
$$A = \ell w$$



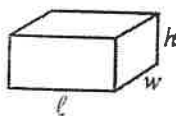
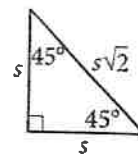
$$A = \frac{1}{2}bh$$



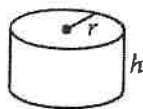
$$c^2 = a^2 + b^2$$



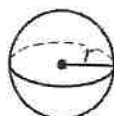
Special Right Triangles



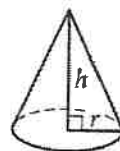
$$V = \ell wh$$



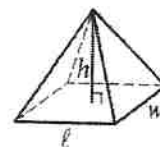
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

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1

If $x > 0$, which of the following is equivalent to $\sqrt[3]{x^8}$?

- I. $(x^8)^{\frac{1}{3}}$
- II. $(x^3)^{\frac{1}{8}}$
- III. $(x^{\frac{1}{3}})^2 \cdot x^2$

- A) I only
- B) I and III only
- C) II only
- D) II and III only

2

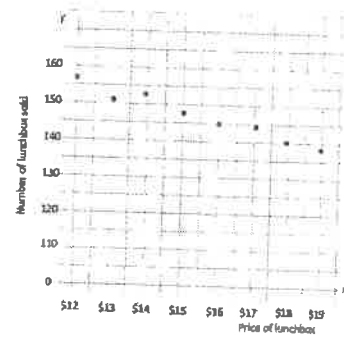
$$8^x \cdot y^2 = 8^{x-1} \cdot y$$

In the equation above, x and y are positive numbers. What is the value of y ?

3

If $3x + 2y = -7$, what is the value of $9x^2 + 12xy + 4y^2$?

4



A certain restaurant sells take-out lunch boxes ranging in price from \$12 to \$19 in increments of \$1. The scatter plot above shows the total number of lunchboxes sold at each price. Approximately how much more revenue did the restaurant get from the lunchboxes it sold priced at \$17 than it did at \$12?

- A) \$400
- B) \$580
- C) \$820
- D) \$1,100

5

A party planner is ordering balloons for a party. The planner wants to have at least 100 balloons in her order, and she will order no more than twice as much red balloons as white balloons. Each cost for red balloon and white balloon is \$1.25 and \$1.50, respectively. If r represents the number of red balloons and w represents the number of white balloons and her budget is \$1,000, which of the following inequalities below best represents this situation?

- A) $\begin{cases} r + w \geq 100 \\ 1.25r + 1.50w \leq 1,000 \\ r \leq 2w \end{cases}$
- B) $\begin{cases} r + w \geq 100 \\ 1.25w + 1.50r \leq 1,000 \\ r \leq 2w \end{cases}$
- C) $\begin{cases} r + w \geq 100 \\ 1.25r + 1.50w \leq 1,000 \\ w \leq 2r \end{cases}$
- D) $\begin{cases} r + w \geq 100 \\ 1.25w + 1.50r \leq 1,000 \\ w \leq 2r \end{cases}$

6

$$k(x + 1)(x - 1) + 3x^2 - 2x = ax^2 + bx + c$$

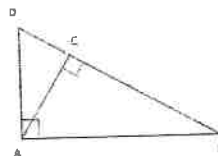
In the equation above, a , b , c , and k are constants. If the equation is true for all values of x , what is the value of $a + b + c$?

7

Jeremiah deposited his allowance \$150 into a bank savings account which earns an annual interest rate of 6%. If he doesn't withdraw any money nor make any additional deposits, how long will it take him, in years, to increase the value of account by at least 40%?

- A) 4
B) 5
C) 6
D) 7

8



Note: Not drawn to scale.

In the right triangle ABD above, the value of $\cos B$ is 0.2 and the length of BC is 5. What is the length of CD?

- A) 5
B) 25
C) 120
D) 125

9

$$V = \frac{4}{3}\pi r^3$$

The volume of sphere is shown above, where r is the radius of the sphere. If the value of the radius is double, what happens to the value of the volume of the sphere?

- A) Double the original value
- B) Triple the original value
- C) Eight times the original value
- D) 27 times the original value

10

An invasive insect was found to have a population of P after t weeks of uninhibited growth.

$$P(t) = C(2)^{\frac{t}{4}}$$

The equation above gives the number of populations, P , of the insects, where t is the number of weeks after the uninhibited growth began, and C is the number of initial population. Which of the following statements is valid based on the equation above?

- A) The population will be increased by 100% every 4 weeks.
- B) The population will be increased by 200% every 4 weeks.
- C) The population will be double every 2 weeks.
- D) The population will be double every week.

11

Jerome usually spends $3\frac{3}{4}$ hours a day working on his math and English homework. It takes him $\frac{1}{2}$ hour to complete one of his math assignments and $\frac{3}{4}$ hour to complete one of his English assignments. If m represents the number of math assignments and e represents the number of English assignments, which of the following best represents this situation?

- A) $\left(m + \frac{1}{2}\right)\left(e + \frac{3}{4}\right) = \frac{15}{4}$
- B) $\left(\frac{m}{2}\right)\left(\frac{3e}{4}\right) = 1$
- C) $\left(\frac{1}{2}m\right)\left(\frac{3}{4}e\right) = \frac{15}{4}$
- D) $\frac{1}{2}m + \frac{3}{4}e = \frac{15}{4}$

12

$$|3x + 1| = |-x|$$

In the equation above, how many solutions to the equation exists?

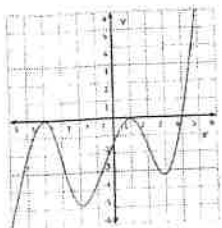
- A) None
- B) One
- C) Two
- D) Three

13

The sum of three numbers is 220. If the largest number is 20 percent more than the sum of the other two numbers, what is the value of the largest number?

- A) 200
- B) 100
- C) 150
- D) 120

14



The graph of function f is shown in the xy -plane, where $y = f(x)$. Which of the following functions could be f ?

- A) $f(x) = (x + 4)(x - 1)^2(x - 4)$
- B) $f(x) = -(x + 4)^2(x - 1)^2(x - 4)$
- C) $f(x) = (x + 4)^2(x - 1)^2(x - 4)$
- D) $f(x) = (x + 4)^2(x - 1)^3(x - 4)$

15

$$t = \sqrt{\frac{k}{4hl}}$$

The experimental equation above gives the estimate time, called a half-life(t), for a certain element to decay 50% of its original amount. k ; the average temperature of soil, h ; the ph. level of the soil, and the humidity index ; l .

Which of the following correctly expresses the humidity index in terms of other variables?

- A) $l = \left(\frac{16th}{k}\right)^2$
- B) $l = \frac{k}{4ht^2}$
- C) $l = \frac{k}{2ht^2}$
- D) $l = \left(\frac{k}{4ht}\right)^2$

16

A parabola of $y = f(x)$ is graphed in the xy -plane, where $f(x) = (x - 1)(x - 4)$. Which of the following intervals contains the x -coordinate of the vertex of the graph of $f(x)$?

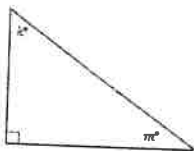
- A) $0 < x < 1$
- B) $1 < x < 3$
- C) $4 < x < 5$
- D) $5 < x < 6$

17

During a special sale, Edward spent a total of d dollars, which includes the discounted price at 25% discount and the sales tax 9.5% on the discounted price, on a certain item. In terms of d , what was the original price of the item?

- A) $\frac{(0.75)(d)}{1.095}$
 B) $(1.095)(d)(0.75)$
 C) $\frac{d}{(1.095)(0.25)}$
 D) $\frac{d}{(1.095)(0.75)}$

18



In the right triangle above, k and m are two acute angles of the right triangle. Which of the following operations is correct for the following expressions based on the figure?

$$\cos(m^\circ) \text{ ______ } \sin(k^\circ)$$

- A) $>$
 B) $<$
 C) $=$
 D) Can't be determined.

19



The graph of $y = f(x) + 2$ is shown above. Which of the following could define f ?

- A) $f(x) = -3^x + 2$
 B) $f(x) = -3^x + 1$
 C) $f(x) = 3^x + 1$
 D) $f(x) = 3^x - 1$

20

The area of circle is increasing at a rate of 15 square inches per hour. Which of the following expressions correctly expresses to this rate in square meters per min? (1 inch = 2.54 centimeters, 1 meter = 100 centimeters)

- A) $\frac{(15)(2.54^2)}{(100^2)(60)}$
 B) $\frac{(15)(100^2)}{(2.54)^2(60)}$
 C) $\frac{(15)(100)^2}{(2.54)^2}$
 D) $\frac{(15)(60)^2}{(2.54)^2(100)}$

21

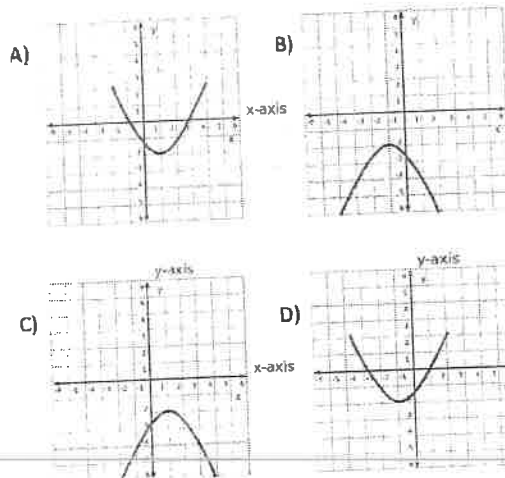
$$\begin{aligned}x^2 + 4x + y^2 - 2y - 4 &= 0 \\ x - k &= 0\end{aligned}$$

In the system of equations above, what is a possible value of k if the system has exactly two real solutions?

- A) 0
- B) 1
- C) 2
- D) 3

22

The quadratic function h is defined by $h(x) = -(x - 1)^2 - 2$. Which of the following could be the graph of $y = h(x)$ shifted 2 units to the left?

**STOP**

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.